

RULE NO. 15

GENERATING FACILITY INTERCONNECTIONS

A. APPLICABILITY

1. This Rule describes the Interconnection, operating and Metering requirements for Generating Facilities intended to be connected to the Utility's electric Distribution System over which the Commission has jurisdiction.
2. This Rule applies only to Generating Facilities with a Maximum Generating Capacity of 20 MW or less unless otherwise required in federal or state law. Generating Facilities with a Maximum Generating Capacity of greater than 20 MW are not subject to this Rule and shall be addressed through the Utility's Large Generator Interconnection Agreement or other appropriate process.
3. Generating Facilities seeking to Interconnect to the Utility's Distribution System and sell a portion of or all their electrical output to the Utility or a third party are not subject to this Rule and shall be addressed through the Utility's Small Generator Interconnection Agreement, Large Generator Interconnection Agreement, Qualified Facility, or other appropriate process, as applicable.
4. The requirements of this Rule apply to Generating Facilities Interconnected with the Utility's Distribution System that will operate in Parallel Operation except as identified in Sections A.2 and A.3. Attachment 1 provides guidance in determining the applicability of this Rule for anticipated common situations (but does not and cannot contain all possible situations).
 - a. EVSE paired with a Generator is subject to this Rule for the Generator only. Charge-only EVs connected to chargers or charging stations are not subject to this Rule.
 - b. V2G EVs and EVSE are subject to this Rule only when configured to discharge to the Utility's Distribution System while in Parallel Operation, including conversions of charge-only equipment to V2G.
5. Producers seeking to connect Generating Facilities to the Utility's Distribution System that will operate in Parallel Operation or Momentary parallel Operation (where a continuous electrical path will be established between the Generating Unit(s) and the Utility's Distribution System at any time, and where the Generating Facility is capable of producing or discharging electricity while connected to or following disconnection ~~from from~~ the Utility's Distribution System), shall file an Application under the requirements of Section E of this Rule to enable the Utility to determine if the connection and proposed operation will result in Parallel Operation, Momentary Parallel Operation, or Neither, and therefore the requirements of this Rule apply or not.
6. The Interconnection of a Net Metering System is subject to this Rule, except as otherwise specified herein, and may require a different Application or Interconnection and Operating Agreement than those applicable to non-Net Metering System Interconnections.
7. Generating Facilities interconnected prior to the effective date of this Rule shall be governed by the requirements of this Rule in effect at the time the Generating Facilities were installed and their existing Interconnection and Operating Agreements, unless otherwise modified by mutual agreement of the Producer and Utility.
8. Capitalized terms used in this Rule, and not defined in the Utility's other tariffs, shall have the meaning ascribed to such terms in Section B of this Rule. The definitions set forth in this Rule shall apply only to this Rule and related tariff riders for Standby Service and may not apply to the Utility's other Rules.
9. Generating Facilities that are intended solely for isolated operation (e.g., standby generators using Break-before-Make transfer equipment) or Momentary Parallel Operation, both in terms of their load and generation characteristics (e.g., Energy Storage Devices), and never operating in Parallel Operation with the Utility's Distribution System, shall not be subject to this Rule. The connection of such Generating Facilities to the Utility's Distribution System shall be addressed under the Utility's Rule 19.

Commented [ZW1]: VGIC appreciates NV Energy adding V2G into Rule 15, given the growing availability of solutions from Ford, GM, Nissan, Kia, Tesla, Blue Bird, Thomas Built, IC Bus, RIDE, Wallbox, dcbel, InCharge, Heliox, Tellus Power Green, and many others expected to become available to customers in the next 12-18 months.

Commented [ZW2]: Clarification question: does this statement imply that this interconnection rule (Rule 15) only applies to generators that operate in a non-export mode, and that for generators that operate in an export mode, there are separate processes/rules?

Relatedly, if NV Energy were to establish a pilot, program, or rate to compensate EV/EVSE for exports to the grid, would those systems be subject to a different interconnection rule?

Commented [ZW3]: Typo

10. Subject to Sections A.4 and A.5, the determination of whether the proposed connection of a Generating Facility to the Utility's Distribution System is subject to the requirements of this Rule or Rule 19 shall be at the discretion of the Utility.
11. Modification of the Utility's Distribution System as a result of a Producer's Application for Interconnection under this Rule, including both Interconnection Facilities and Distribution System improvements which may solely result from either the load or generation characteristics of a Generating Facility, or the Generating Facility's auxiliary load, shall be as set forth in Rule 9 and a separate Rule 3 application shall not be required as a result of a Producer's Application for Interconnection under this Rule in this instance.
12. Subject to the provisions of this Rule, the Utility shall allow the Interconnection of Generating Facilities with its electric Distribution System.

B. DEFINITIONS AND ACRONYMS

The definitions set forth in this Section B are applicable only to this Rule, its corresponding agreements and rate schedules that apply to Standby Service.

AC: Alternating Current.

Active Anti-Islanding Scheme: A control scheme installed with the Generating Facility that senses and prevents the formation of an Unintentional Island.

Active Affordability Covenant: A covenant, condition, or restriction contained in a deed, contract, or other legal instrument which establishes a period of affordability and limits rent and utility charges for low-income households.

Advanced Inverter (a.k.a., Smart Inverter): A Generating Facility's inverter capable of performing functions that when activated can autonomously contribute to grid support during excursions from normal operating voltage and frequency system conditions by providing dynamic reactive/real power support, voltage and frequency ride-through, ramp rate controls, communication systems with ability to accept external commands and other functions. Such inverters can either be Grid Following or Grid Forming but must be grid interactive.

ANSI: American National Standards Institute.

Applicant: The entity applying for Interconnection under the provisions of this Rule. The Applicant may or may not be the Producer or Customer.

Application: Documentation submitted to the Utility by the potential Applicant that contains sufficient information necessary to begin the Interconnection process. This also includes instances where a modification(s) to an existing Interconnection's equipment or operation take place necessitating the Utility to examine the potential effect(s) of the modification(s).

AHJ: Authority Having Jurisdiction

Break-before-Make: A type of electrical transfer method where the connection to one power source is completely disconnected before a new connection to another source is established.

Certification; Certified; Certificate: The documented results of a successful Certification Testing.

Certification Test(ing)(s): A test(ing) adopted by the Utility that verifies conformance of certain equipment with Commission-approved performance standards in order to be classified as Certified Equipment. Certification Tests are normally performed by a Nationally Recognized Testing Laboratory such as the Underwriter's Laboratory. Attachment 2 to this Rule outlines such Certification Testing.

Certified Equipment: Equipment used in a Generating Facility that has passed the Certification Test.

Commission: The Public Utilities Commission of Nevada

Commissioning Test(ing)(s): A test performed during the commissioning of all or part of a Generating Facility system to achieve one or more of the following: verify specific aspects of its performance; calibrate its instrumentation; and establish instrument or Protective Function set-points.

Customer(s): is defined in Rule 1.

Customer Microgrid: A Microgrid that uses only non-Utility infrastructure on the customer side of the Point of Common Coupling, including distribution lines, Generating Facilities, and related equipment, to meet its interconnected loads.

DC: Direct Current.

Dedicated Transformer; Dedicated Distribution Transformer: A transformer that provides electric service to a single Customer. The Customer may or may not have a Generating Facility.

Detailed Review: Part of the technical review process as determined by Section F.6 of this Rule. The Detailed Review includes the Utility's performance of an Interconnection Study that establishes the requirements for Interconnection of a Generator or Generating Facility and may be performed as a result of unremedied failure of any of the technical review Screens or by request of the Applicant.

Distributed Energy Resource(s)(DER)(s): Distributed generation systems, energy efficiency, energy storage (including Energy Storage Devices), Electric Vehicles and demand-response technologies, which may be located on the Utility side or the Customer side of the Meter of a Utility Customer. This definition conforms to the definition of distributed resources in NAC 704.90583 and shall be interpreted as meant to conform to future versions of that regulation unless otherwise stated in this Rule.

Distribution System: The Utility's electrical system consisting of all electrical wires, equipment, and other facilities owned or provided by the Utility by which the Utility provides distribution service, including High Voltage Distribution service as defined in Rule 9, to its Customers.

Electric Vehicle Supply Equipment (EVSE): The physical and electrical equipment and components and software associated with an Electric Vehicle charger or charging station that facilitates the charging or discharging of Electric Vehicles.

Emergency: An actual or imminent condition or situation, which jeopardizes the Utility's Distribution System integrity.

Energy Storage Device (ESD): A device that captures energy produced at one time, stores that energy for a period of time, and delivers that energy as electricity for use at a future time. For purposes of this Rule only and only where applicable within this Rule, an ESD can be considered a Generator. While an ESD may be paired with a Net Metering System, it does not qualify to be considered as a Net Metering System, either as a stand-alone installation or when paired with a Net Metering System. For purposes of this Rule only and only where applicable within this Rule, a Vehicle-to-Grid Electric Vehicle, including its EVSE, can be considered an ESD.

Enter Service: The beginning of Parallel Operation of a Generator with the Utility's energized Distribution System.

EV: Electric Vehicle. An EV may be operated in a charging only (load-only) mode or in a discharging (Vehicle-to-Grid) mode.

Export; Exported; Exporting: The transfer of electricity from a Generating Unit or Generating Facility to the Utility across the Point of Common Coupling. This may include scenarios where Limited Export or Inadvertent Export could occur, or in association with Limited Generation Profile operation.

Fast Track Review: Part of the technical review process as determined by Section F.3 of this Rule. The Fast Track Review may result in approval of the Interconnection, the Utility performing Supplemental Review, or an Applicant's options meeting if requested by the Applicant to discuss any failed Screens.

FERC: Federal Energy Regulatory Commission.

Field Testing: Testing performed in the field to determine whether equipment meets the Utility's requirements for safe and reliable Interconnection.

Generating Facility: All Generating Units that are included in a Generating Facility Interconnection and Operating Agreement.

Generator or Generating Unit: A device that converts mechanical, chemical or solar energy into electrical energy, including all of its protective and control functions and structural appurtenances. One or more Generators comprise a Generating Facility.

Grid Following Inverter: A type of Advanced Inverter that can synchronize its output voltage and frequency with the existing grid voltage and frequency.

Grid Forming Inverter: A type of Advanced Inverter that can create and regulate grid voltage and frequency and can operate independently or in coordination with other sources.

Gross Nameplate Rating: The gross generating AC capacity (kW) of a Generating Unit or the total of the gross generating capacities (kW) of the Generating Units comprising a Generating Facility as designated by the manufacturer(s) of the Generating Facility(s). For an EV, the gross nameplate rating shall be determined by the gross nameplate discharge power of the inverter (kW), and not the battery energy capacity (kWh).

Host Load: Electrical power that is consumed by the Customer at the property on which the Generating Facility is located. This includes the consumption of electrical power of the Generating Unit(s) comprising the Generating Facility.

Hosting Capacity: The incremental amount of DER Maximum Generating Capacity or load that can be accommodated (i.e., is available) on a particular feeder section of the Utility's electric Distribution System at a given time under existing and forecasted grid conditions and operations without adversely impacting safety, power quality, reliability, or other operational criteria. Such capacity may be in terms of generation hosting capacity, photovoltaic hosting capacity, or load hosting capacity.

Hybrid Microgrid: A Microgrid that uses Utility and non-Utility infrastructure on the Microgrid's side of the Microgrid Interconnection Device, including distribution lines, Generating Facilities, and related equipment, to meet its interconnected loads.

IEEE: Institute of Electrical and Electronics Engineers.

Inadvertent Export: The unplanned, uncompensated transfer of active power from a Generator or Generating Facility to the Utility's Distribution System across the Point of Common Coupling, limited as established in this Rule.

In-rush Current: The current drawn by the Generating Facility during startup.

Intentional Island: The planned creation of an Island, usually following a loss of a portion of the Utility's Distribution System, with the approval of the Utility.

Interconnection and Operating Agreement (IOA): An agreement between the Utility and the Producer that gives each certain rights and obligations to effect, operate, modify, or end Interconnection.

Interconnection; (Interconnected): The physical connection of a Generating Facility whereby a continuous electrical path is established between the Generating Unit(s) and the Utility's Distribution System for more than fifteen (15) seconds at any time and during any planned operating mode of the Generating Unit(s) in accordance with the requirements of this Rule so that Parallel Operation with the Utility's Distribution System can occur (or has occurred if Interconnection Authorization (if applicable) and Permission to Operate has been provided by the Utility).

Interconnection Facilities: The electrical wires, switches and related equipment that interconnect a Generating Facility to the Utility's Distribution System.

Interconnection Study: A study to establish the requirements for Interconnection of a Generator or Generating Facility.

Interconnection Study Agreement: An agreement between the Utility and the Applicant that gives each certain rights and obligations regarding the Utility's performance of an Interconnection Study.

Island; Islanded; Islanding: A condition on the Utility's Distribution System in which one or more Generating Facilities deliver power to Customers using a portion of the Utility's Distribution System that is electrically isolated from the remainder of the Utility's Distribution System. A Hybrid Microgrid is a form of an Island.

kVA: Kilovolt-ampere.

kVAR: Kilovolt-ampere reactive.

kW: Kilowatt(s).

Commented [ZW4]: Although this is likely intuitive to any users of this Rule, VGIC recommends explicitly clarifying this to reduce confusion.

kWh: Kilowatt-hour(s).

Limited Export(ing): The limiting of the transfer of electrical energy from a Generating Unit or Generating Facility to the Utility across the Point of Common Coupling by the use of any of the methods or operating modes noted in Section G.2. This may include scenarios where Inadvertent Export could occur, or in association with Limited Generation Profile operation.

Limited Generation Profile: An energy generation schedule or control profile implemented by a system or equipment configuration that specifies the maximum amount of electricity a Generator may produce, or an ESD or Vehicle-to-Grid EV may discharge, at specific times of the day and/or different times of the year. A Limited Generation Profile may be executed by a Certified Power Control System or energy management system with autonomous operational capabilities, or other control architectures (e.g., relay-based schemes) deemed acceptable and approved by the Utility at its discretion to reliably enforce specified energy generation or discharge limits.

Line Section: That portion of the Utility's Distribution System connected to a Customer bounded by automatic sectionalizing devices or the end of the line.

Master Microgrid Controller: An intelligent Power Control System designed to manage and automate the operation of a Microgrid and has the capability to monitor, control, and obtain dynamic feedback from all the individual components that make up a Microgrid.

Material Modification: A change or replacement of generating equipment that is not a like-kind substitution in size, ratings, impedances, efficiencies, or capabilities of the equipment specified in the original Application or Interconnection and Operating Agreement. Table 2 in Section E.3 of this Rule provides guidance regarding what changes to the existing circumstances of a Generating Unit's or Generating Facility's equipment or operation constitute a Material Modification.

Maximum Generating Capacity: The maximum electrical power output that a Generating Facility is capable of producing that is equal to the total Net Nameplate Rating of a Generating Facility when no limiting of its generating capacity is employed or is an amount less than the total Net Nameplate Rating when the generating capacity of a Generator or Generating Facility is limited (e.g., through the use of a Power Control System, power relay(s), Protective Function(s), Limited Generation Profile operation, or as determined by other similar device settings or adjustments), as determined by the limiting method. This capacity may be specified by the Applicant in the Application but shall be specified in the Net Metering System agreement or Interconnection and Operating Agreement and may reflect the total amount of generation output including ESD or Vehicle-to-Grid EV discharge, if applicable. For Non Exporting systems, Inadvertent Export is not included.

Meter: is defined in Rule 1.

Metering Equipment: All equipment, hardware, software including Meter cabinets, conduit, etc. that is necessary for Metering.

Metering: The measurement of electrical power flow in kW and/or kWh, and, if necessary, kVAR, at a point, and its display to the Utility, as required by this Rule.

Microgrid: A self-sufficient energy system that serves a discrete geographic footprint, is comprised of a group of Interconnected loads and DERs that acts as a single controllable entity with respect to the grid, can operate either in Parallel Operation or Islanded, and has the defining characteristics as noted in Section G.10 of this Rule.

Microgrid Interconnection Device: A device that facilitates electrical separation of a Microgrid from the Utility's Distribution System. This device may provide the function of a normal paralleling device to reconnect the Microgrid with the Utility's Distribution System.

Momentary Parallel Operation: The Interconnection of a Generating Facility to the Utility's Distribution System for fifteen seconds (900 cycles) or less.

MW: Megawatt(s).

NAC: Nevada Administrative Code.

Nationally Recognized Testing Laboratory (NRTL): A laboratory accredited to perform the Certification Testing requirements under this Rule.

NEC: The National Electric Code.

Net Energy Metering: Metering for the receipt and delivery of electricity between the Producer and the Utility pursuant to Nevada Revised Statutes 704.769 and the Utility's Net Energy Metering Tariff Schedules, as may be amended from time to time.

Net Generation Metering: The Metering of the net electrical energy output in kW and kWh from a given Generating Facility. This may also be the measurement of the difference between the total electrical energy produced by a Generating Unit and the electrical energy consumed by the auxiliary equipment necessary to operate the Generating Unit. For a Generating Unit with no Host Load, Metering that is located at the Point of Common Coupling. For a Generating Unit with Host Load, Metering that is located at the Generating Unit bus after the point of auxiliary load(s) and prior to serving Host Load.

Net Metering Systems: Those systems operating in Parallel Operation with the Utility as set forth in NRS 704.766 to 704.775 and the Utility's Net Energy Metering Tariff Schedules, as may be amended from time to time. A Net Metering System is defined in NRS 704.771. The addition of an ESD must not result in a violation of any limitations placed on Net Metering Systems in NRS 704.771.

Net Nameplate Rating: The gross AC generating capacity of a Generating Unit or the total of the gross AC generating capacity of the Generating Units comprising a Generating Facility as designated by the manufacturer(s) of the Generating Unit(s) minus the consumption of electrical power of the Generating Unit(s). The Net Nameplate Rating shall be specified in the Net Metering System agreement or Interconnection and Operating Agreement.

Network Service: More than one electrical feeder providing distribution service at a Point of Common Coupling.

NFPA: National Fire Protection Association.

Non-Export(ing): Designed to prevent or limit the transfer of electrical energy from a Generating Unit or Generating Facility across the Point of Common Coupling to the Utility by any of the options in Section G.2. of this Rule. Non-Exporting includes scenarios where Inadvertent Export could occur.

Non-Islanding: Designed to detect and disconnect from a stable Unintentional Island with matched load and generation. Reliance solely on under/over voltage and frequency trip is not considered sufficient to qualify as Non-Islanding.

NRS: Nevada Revised Statutes.

Optional Pre-application Report: A Utility-provided report that certain potential Applicants have the option to request to acquire information about the Utility's Distribution System conditions at their proposed Point of Interconnection prior to submitting a full Application for Interconnection.

Optional Pre-application Report Request Form: The optional form that must be completed by a potential Applicant to request that the Utility provide a Pre-application Report containing information about the Utility's Distribution System conditions at their proposed Point of Interconnection. **Parallel Operation:** The simultaneous operation of a Generating Unit or Generating Facility, including power delivered to or received by the Utility's Distribution System, power delivered from the Utility's Distribution System (e.g., in the case of the charging of an ESD from the Utility's Distribution System), or an idle state, while Interconnected.

Periodic Test(ing): A test performed on part or all of a Generating Facility at pre-determined time or operational intervals to achieve one or more of the following: 1) Verify specific aspects of its performance, 2) Calibrate instrumentation, 3) Verify and re-establish instrument or Protective Function set-points.

Permission to Operate (PTO): The Utility's express written permission authorizing the Generating Unit or Generating Facility to commence Parallel Operation. This express written permission is separate from and shall occur following Interconnection Authorization if such Interconnection Authorization is granted by the Utility.

Point of Common Coupling Metering: Metering located at the Point of Common Coupling. This is the same Metering as Net Generation Metering for Generating Facilities with no Host Load.

Point of Common Coupling (PCC): The transfer point for electricity between the electrical conductors of the Utility and the electrical conductors of the Producer.

Point of Interconnection (POI): The electrical transfer point between a Generating Facility and the Utility's existing or planned Distribution System. This may or may not be coincident with the Point of Common Coupling.

Power Control System (PCS): A system that limits or controls the generation output of Generators or a Generating Facility, including ESDs or V2G EVs, or the load requirement of such technologies. Such a system may be Advanced Inverter-based or based upon other Certified technologies as approved by the Utility on a case-by-case basis at its discretion and may establish or manage a Limited Generation Profile or Limited Export profile. A Certified PCS shall conform to UL 3141 only if and when such standard is fully approved, and then, only after its implementation date.

Producer: The entity that executes an Interconnection and Operating Agreement with the Utility. The Producer may or may not own and operate the Generating Facility but is responsible for the rights and obligations related to the Interconnection and Operating Agreement.

Production Test(ing): A test performed on each device coming off the production line to verify certain aspects of its performance.

Protective Function(s): The equipment, hardware and/or software in a Generating Facility (whether discrete or integrated with other functions) or on the Utility Distribution System whose purpose is to protect against Unsafe Operating Conditions.

Prudent Electrical Practices: Those practices, methods, and equipment, as changed from time to time, that are commonly used in prudent electrical engineering and operations to design and operate electric equipment lawfully and with safety, dependability, efficiency, and economy.

Qualified Low-Income Residential Building: As defined in Nevada Assembly Bill 458 (2025), referenced here only to establish eligibility for Interconnection.

Qualifying Facility: is defined in Rule 1.

Qualified Multifamily Affordable Housing Property (QMAHP): As defined in Nevada Assembly Bill 458 (2025), referenced here only to establish eligibility for Interconnection.

Reference Point of Applicability (RPA): A location proximate to the Generating Facility where the Interconnection and interoperability performance requirements as specified by IEEE 1547-2018 apply. This location shall be specified by the Applicant in the Application and determined by mutual agreement between the Applicant and the Utility.

Relevant Minimum Load: The lowest Line Section or substation load either measured or reasonably estimated by the Utility coincident with the Generating Facility's production. For inverter-based solar photovoltaic Generators with no paired ESD, the daytime minimum load shall be used (10 AM to 4 PM for fixed panel systems and 8 AM to 6 PM for systems utilizing tracking).

Rule: is defined in Rule 1.

Scheduled Operation Date: The date specified in the Generating Facility's Interconnection and Operating Agreement when the Generating Facility is, by the Producer's estimate, expected to begin initial operation.

Screen(s): The aspects of the technical review process that gauge a Generator's or Generating Facility's impact on the electrical Distribution System of the Utility and adherence to certain technical requirements and determine if a Generator or Generating Facility Application passes or fails Simplified Review, Fast Track Review, Supplemental Review, or if Detailed Review is required.

Secondary Network: A network supplied by several primary feeders suitably interlaced through the area in order to achieve acceptable loading of the transformers under emergency conditions and to provide a system of extremely high service reliability. Secondary networks usually operate at 600 V or lower.

Service(s): is defined in Rule 1.

Simplified Review: Part of the technical review process as determined by Section F.2 of this Rule. The Simplified Review may result in approval of the Interconnection or an Applicant's options meeting to discuss any failed Screens.

Single Line Diagram; Single Line Drawing: A schematic drawing, showing the major electric switchgear, Protective Function devices, wires, Generators, transformers and other devices, providing sufficient detail to communicate to a qualified engineer the essential design and safety of the system being considered.

Short Circuit Contribution Ratio (SCCR): The ratio of the Generating Facility's short circuit contribution to the Utility's short circuit contribution for a three-phase fault at the high voltage side of the distribution transformer connecting the Generating Facility to the Utility's Distribution System.

Short Circuit Interrupting Capability: The maximum fault current that can be safely interrupted by a protective device.

Solar-Powered Affordable Housing System: As defined in AB 458 (2025). For purposes of this Rule, such systems are treated as Net Metering Systems.

Special Facilities: Special Facilities are: (a) facilities requested by the Producer which are in addition to or in substitution for standard facilities which the Utility would normally provide for delivery of service at one point, through one Meter, at one voltage class under its Tariff Schedules, or (b) a pro rata portion of the facilities requested by an Applicant, allocated for the sole use of such Applicant, which would not normally be allocated for such sole use. Unless otherwise provided by the Utility's filed Tariff Schedules, Special Facilities will be installed, owned and maintained or allocated by the Utility as an accommodation to the Producer only if acceptable for operation by the Utility and the reliability of service to the Utility's other customers is not impaired.

Stability: The return to normalcy of the Utility's Distribution System following a disturbance. Stabilization is usually measured as a time period during which voltage and frequency are within acceptable ranges.

Standby Service: is defined in Rule 1.

Starting Voltage Drop: The percentage voltage drop at a specified point resulting from In-rush Current. It can also be expressed in volts on a particular base voltage, (e.g., 6 volts on a 120-volt base, yielding a 5 percent drop).

Supplemental Review: Part of the technical review process as determined by Section F.4 of this Rule. The Supplemental Review may result in approval of the Interconnection or a meeting between the Utility and the Applicant to discuss the Applicant's options.

Synchronization (Synchronizing): The requirement that at the time of Interconnection of a Generator, in accordance with the synchronization performance requirements of IEEE 1547-2018, Clause 4.1.3 and Table 5.

System Integrity: The condition under which a Distribution System is deemed safe and can reliably perform its intended functions in accordance with the safety and reliability rules of the Utility.

Tariff Schedule(s): is defined in Rule 1.

TCP/IP: Transmission Control Protocol/Internet Protocol, which is a suite of communication protocols used to interconnect network devices on the internet.

Telemetry: The electrical or electronic transmittal of Metering data on a real-time basis to the Utility.

Temporary Interconnection Authorization (TIA): If required by the Utility for certain Interconnections, the express written permission by the Utility authorizing the Interconnection of a Generating Facility for the limited purposes of inspection, testing, coordination, and commissioning. TIA is not PTO, which authorizes Parallel Operation, and may be issued prior to the Utility's issuance of PTO.

Transfer Trip (Capability): A Protective Function that trips a Generating Facility remotely by means of an automated communications link controlled by Utility.

Type Test(ing): A test performed on a sample of a particular model of a device to verify specific aspects of its design, construction and performance.

UL: Underwriters Laboratory, Inc.

Unauthorized Interconnection: The physical connection of a Generating Facility to the Utility's Distribution System, or the commencement of Parallel Operation, without first obtaining written Temporary Interconnection Authorization or PTO from the Utility in accordance with this Rule. An Unauthorized Interconnection also includes any circumstance in which a Generating Facility is energized and capable of Exporting power to the Utility's Distribution System prior to the Utility's express written authorization.

Unintentional Island: The unplanned creation of an Island, usually following a loss of a portion of the Utility's Distribution System, without the approval of the Utility.

Unsafe Operating Conditions: Conditions that, if left uncorrected, could result in harm to personnel, damage to equipment, loss of System Integrity or operation outside pre-established parameters required by the Interconnection and Operating Agreement.

Visible-Break Isolation Device: An electrical switching device that can separate the Generating Facility from Utility's Distribution System and is designed to allow visible verification that separation has been accomplished. This requirement can be met by opening the enclosure to observe the contact separation.

Vehicle-to-Grid (V2G): The bidirectional transfer of electricity between an EV and the Utility's Distribution System when enabled through Certified bidirectional EVSE and operating in Parallel Operation. V2G operation may include charging from the Utility's Distribution System, discharging to serve all or a portion of onsite Host Load, and/or Export to the Utility's Distribution System. For purposes of this Rule, a V2G EV is considered an ESD.

C. APPLICABLE CODES AND STANDARDS

As applicable, Interconnections shall conform to the most recent approved versions of the codes and standards listed below (which may be a more recent year than indicated below) on the date the Application is accepted as complete and correct by the Utility, or successor standards:

ANSI C84.1-2020: Electric Power Systems and Equipment - Voltage Ratings (60 Hertz).

IEEE C2-2023: 2023 National Electric Safety Code® (NESC®).

IEEE C62.92.6™-2017: IEEE Guide for Application of Neutral Grounding in Electrical Utility Systems, Part VI - Systems Supplied by Current Regulated Sources.

IEEE Std 1453™-2022: IEEE Standard for Measurement and Limits of Voltage Fluctuations and Associated Light Flicker on AC Power Systems.

IEEE Std 1547™-2018: IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces, as amended by IEEE 1547a™-2020, including use of IEEE 1547.1™-2020 testing protocols to establish conformity.

IEEE Std 1547.1™-2020: IEEE Standard Conformance Test Procedures for Equipment Interconnecting Distributed Energy Resources with Electric Power Systems and Associated Interfaces.

IEEE Std 1547.2™-2008: IEEE Application Guide for IEEE Std 1547™, IEEE Standard for Interconnecting Distributed Resources with Electric Power Systems.

IEEE Std 1547.3™-2023: IEEE Guide for Cybersecurity of Distributed Energy Resources Interconnected with Electric Power Systems.

IEEE Std. 1547.4™-2011: IEEE Guide for Design, Operation, and Integration of Distributed Resource Island Systems with Electric Power Systems.

IEEE Std 1547.6™-2011: IEEE recommended practice for interconnecting distributed resources with electric power systems distribution secondary networks.

IEEE Std 1547.7™-2013: IEEE Guide for Conducting Distribution Impact Studies for Distributed Resource Interconnection.

IEEE Std 1815™-2012: IEEE Standard for Electric Power Systems Communications—Distributed Network Protocol (DNP3).

IEEE Std 2030.5™-2023: IEEE Standard for Smart Energy Profile 2.0 Application Protocol (SEP 2.0).

IEEE Std 2030.2™-2015: IEEE Guide for the Interoperability of Energy Storage Systems Integrated with the Electric Power Infrastructure.

NFPA 70®: National Electric Code® 2017 Edition (as modified by any local code amendments).

SAE J2847/3: Communication for Plug-in Vehicles as a Distributed Energy Source

SAE J3072: Interconnection Requirements for Onboard, Grid Support Inverter Systems.

UL 1741/UL 1741 SA/UL 1741 SB: STANDARD FOR SAFETY Inverters, Converters, Controllers and Interconnection System Equipment for Use with Distributed Energy Resources.

UL 1741 SC: BEVSE/ISE for EV AC V2G applications

UL 1741 CRD: Support UL 9741 Path for Matched Pairing of BEVSE and EV

UL 3141: Outline of Investigation for Power Control Systems (only following this standard's full approval and implementation date).

UL 9741: STANDARD FOR SAFETY Electric Vehicle Power Export Equipment (EVPE)

For Consideration:

UL 9540: ESD system safety and fire code compliance

IEEE 2030.7 / 2030.8: ESD functional specs for advanced control and DERMS integration.

D. **RIGHTS AND OBLIGATIONS**

1. A Producer shall execute an Interconnection and Operating Agreement with the Utility governing the Interconnection and operation of Generating Facilities.
 - a. Pro forma agreements establishing the terms and conditions for Interconnections and operation with the Utility's facilities for each applicable class shall be provided to the Producer by the Utility. Such agreements may be modified by mutual agreement as necessary to address specific Interconnection requirements existent at the time of the execution of the agreement.
 - b. Producers having agreements executed prior to the effective date of this Rule that govern Interconnection and Parallel Operation with the Utility's facilities shall be governed by the provisions of those existing agreements.
 - c. The agreement shall include any necessary requirements for communications and communications facilities between the Utility and the Producer.
2. A Producer requiring other Services not addressed in this Rule may acquire such Services in accordance with the Utility's Tariff Schedules governing the provision of such Services. Specific examples of other such Services include but are not limited to: Supplemental Service for Customer load in excess of what is normally produced by the generating equipment, Backup Service for unscheduled outages of Generating Facilities, and Maintenance Backup Service for scheduled maintenance outages of Generating Facilities. Such Services are outside of the scope of this Rule.
3. Interconnection described hereunder shall not provide a Producer with any rights to use the Utility's system for the transmission, distribution, or wheeling of electric power, nor does Interconnection described hereunder limit those rights. Such rights to the use of the Utility's system must be addressed in separate agreements.

Commented [ZW5]: VGIC recommends this be removed, as communication protocols between EV and EVSE are typically out of scope for utility interconnection rules. Both AC pathways (UL 1741 SC and UL 1741 CRD) ensure compliance with IEEE 1547-2018, and IEEE 1547-2018 requires speaking either IEEE 2030.5, DNP3, or SunSpec Modbus. With this in mind, it is not necessary to define *additional* communications requirements. Doing so may impose unnecessary burden on equipment providers.

Commented [ZW6]: Anticipated February 2026 publication. Per UL: "Clearly defines requirements and compliance criteria; interacts/communicates with EV via SAE J3072; monitor EV V2G AC export grid interactions in accordance with local interconnection requirements; cease EV grid export for safety and grid performance (prevent hazardous overvoltage conditions for local equipment, noncompliance with local grid requirements); refer to and make use of existing standards (UL 1741 and UL 1741 SB, UL 2231-1 and UL 2231-2, and UL 2594)." VGIC has sent Yvette Sanders a recent UL presentation detailing this pathway.

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Commented [ZW7]: Per UL: "Published UL 1741 Standard requires compliance with IEEE 1547-2018 and IEEE 1547.1-2020 to evaluate a single-piece and multipiece DER units and DER systems. Some of the corresponding IEEE 1547 and IEEE 1547.1 terminology and requirements for multipiece DER variations were not included in the previously published UL 1741 text, so the CRD is intended to clarify those requirements: (a) grid support DER unit, (b) grid support DER system, (c) grid support composite DER, and (d) partially compliance grid support DER."

Using both the upcoming UL 1741 CRD and UL 9741, the BEVSE and EV pair is evaluated and certified as a fully compliant DER system to comply with existing IEEE 1547-2018 and IEEE 1547.1-2020 requirements and definitions."

VGIC has sent Yvette Sanders a recent UL presentation detailing this pathway.

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4. Interconnection described hereunder does not require the Utility to purchase the output of a Generating Facility. Such purchase arrangements must be addressed in separate agreements.
5. A Producer shall ascertain and comply with all applicable Commission-approved tariffs of the Utility, applicable FERC-approved rules, tariffs, and regulations, and any local, state or federal law, statute or regulation that applies to the design, siting, construction, installation, operation, or any other aspect of the Generating Facilities and Interconnection Facilities.
6. The Utility shall have the right to review the design of a Producer's Generating Facility and Interconnection Facilities and to inspect a Producer's Generating and/or Interconnection Facilities prior to the provision of Interconnection Authorization (if applicable), PTO and the commencement of Parallel Operation with the Utility's Distribution System. The Utility may require a Producer to make modifications as necessary to comply with the requirements of this Rule. The Utility's review and provision of Interconnection Authorization (if applicable) and/or PTO authorizing Parallel Operation shall not be construed as confirming or endorsing the Producer's design or warranting the Generating and/or Interconnection Facilities' safety, durability or reliability. The Utility shall not, by reason of such review or lack of review, be responsible for the strength, adequacy, or capacity of such equipment.
7. Producers shall not begin Parallel Operation with the Utility's Distribution System, with the exception of the provisions allowable under Section E.2 and any operation solely required to facilitate proper testing and inspection of the Generating Facility in accordance with Section J, until their Interconnection Facilities have been inspected by the Utility and PTO is provided by the Utility to the Producer. Such approval may be withheld for noncompliance with the requirements of this Rule or any of the Utility's applicable tariffs.
8. The Utility may immediately disconnect any Generating Facility that is found to have engaged in an Unauthorized Interconnection. The Producer may also be subject to additional charges, corrective requirements, or other remedies as may be contained in the Net Metering System agreement, Interconnection and Operating Agreement, or as authorized by the Commission.
9. A Producer's Generating Facility and Interconnection Facilities shall be reasonably accessible to Utility personnel as necessary for the Utility to perform its duties and exercise its rights under its tariffs filed with and approved by the Commission, and any agreement between the Utility and the Producer.
10. Any information pertaining to Generating and/or Interconnection Facilities provided to the Utility by a Producer shall be treated by the Utility in a confidential manner.
11. A Producer shall operate and maintain its Generating Facility and Interconnection Facilities in accordance with Prudent Electrical Practices and shall maintain compliance with Commission adopted standards for the Producer's particular Generating Facility and Interconnection Facilities. Said standards shall be those in effect at the time a Producer executes the Interconnection and Operating Agreement with the Utility.
12. The Utility may limit the operation and/or disconnect or require the disconnection of a Producer's Generating Facility from Utility's Distribution System at any time, with or without notice, in the event of an Emergency or to correct Unsafe Operating Conditions. The Utility may also limit the operation and/or disconnect or require the disconnection of a Producer's Generating Facility from the Utility's Distribution System upon the provision of reasonable notice:
 - a. to allow for routine maintenance, repairs or modifications to the Utility's Distribution System;
 - b. for cybersecurity-related concerns;
 - c. upon the Utility's determination that a Producer's Generating Facility is not in compliance with this Rule; or
 - d. upon termination of the Interconnection and Operating Agreement.

The Producer and the Utility shall make good faith efforts to restore the Interconnection of the Generating Facility in a timely manner following a temporary limiting of operation or disconnection.

13. During Parallel Operation, the Producer shall comply with all operational direction of the Utility at the time given with such direction subject to any conditions that the Producer and the Utility may mutually agree to incorporate in the Interconnection and Operating Agreement. Any dispute regarding such direction shall be handled after the fact in accordance with the dispute resolution procedures in Section K, below.
14. The operation of the Generating Facility shall not reduce the quality of service described in Rule 2 to other Customers and shall conform with the Utility's design standards. If the Generating Facility causes service interference, the Utility shall give notice and provide reasonable time for the Producer to take corrective action.
15. When an Interconnection Study is required pursuant to Section F.6, the Producer shall be responsible for paying to the Utility all reasonable costs incurred by the Utility in performing such a study unless such Parallel Operation has been requested by the Utility.

E. APPLICATION FOR INTERCONNECTION

1. Optional Pre-application Report
 - a. A potential Applicant may request an optional Pre-application Report for a specific site to obtain information about the Utility's Distribution System conditions at the proposed POI prior to submitting an Application for Interconnection by completing and submitting to the Utility a Pre-application Report Request Form and a non-refundable fee of \$300.
 - b. The Utility shall make the Pre-application Report Request Form publicly-accessible either as an online form, electronic document transferable through e-mail, or other format convenient to potential Applicants.
 - c. Upon receipt of a completed Pre-application Report Request Form and the non-refundable fee of \$300, absent any extraordinary circumstances, the Utility shall provide the requested data to the potential Applicant within ten (10) business days, making reasonable and good faith efforts to provide all available data requested. However, the Utility shall be obligated to provide existing data relevant to the request only and shall not be obligated to provide data it does not possess at the time of the request nor to conduct a study or other analysis to generate such data. The Utility shall indicate on the form which data it cannot provide at the time of the request.
 - d. None of the data provided by the Utility in an optional Pre-application Report shall in any way imply or guarantee that a potential Applicant's Interconnection can be completed without impacts to the Distribution System or the need for Distribution System Improvements.
2. Application Process
 - a. Prior to receiving an Application, if requested by a potential Applicant and upon such request, the Utility shall provide information and documents (such as the pro forma Interconnection and Operating Agreement and the Application, technical requirements, specifications, listing of Certified Equipment, application fee information, applicable rate schedules and Metering requirements). Unless otherwise agreed upon and absent any extraordinary circumstances, the Utility shall normally send all such information to an Applicant within five (5) business days following the initial request from the potential Applicant.
 - (1) For Net Metering Systems, the Utility shall send a description of the procedures by which a Customer may Interconnect with the Utility's Distribution System under its Net Energy Metering Tariff Schedules and a copy of the standard Net Metering System Interconnection and Operating Agreement and Application form.

- b. All Applicants shall be required to complete and submit an Application for Interconnection to the Utility and supply any relevant additional information requested by the Utility. The submission must include the completed Application and any applicable fee for processing the Application and performing any applicable technical review to be completed by the Utility pursuant to Section F.
- (1) An Application for Interconnection of a Solar-Powered Affordable Housing System must include supporting documentation demonstrating that the system is located on a Qualified Multifamily Affordable Housing Property, including a declaration of Active Affordability Covenants or a verification letter from a funding authority.
- c. The Application processing fee shall vary based on the proposed Generating Facility that will be Interconnected as indicated in Table 1 below:

TABLE 1

Type of Interconnection	Technical Processing Fee	Interconnection Study Fee
Net Metering Systems	\$189*	None
QMAHP	\$XX	\$XX
All Others	Actual Costs Incurred	Actual Costs Incurred

* One fee for Simplified, Fast Track, and Supplemental Reviews.

- e. d. The Applicant may propose, and the Utility may negotiate, specific costs for processing non-standard installations such as multi-units, multi-sites, or otherwise as conditions warrant. The costs in Table 1 may not apply under such circumstances. The Utility shall establish an individual representative as the single point of contact for the Applicant but may allocate responsibilities among its staff to best coordinate the Interconnection of an Applicant's Generating Facility.
- f. The Utility shall record the date and time it first receives an Application.
- g. Within ten (10) business days of first receiving an Application, the Utility shall acknowledge its receipt, complete its first review of the Application, and inform the Applicant if the Application has been completed adequately and has been accepted by the Utility.
- (1) For Net Metering Systems, the Utility shall send a standard Net Metering System Interconnection and Operating Agreement to the Applicant for signature (if the Applicant has not already been sent such agreement) if the data provided in the Application is complete and complies with the technical requirements of NRS 704.774.
- h. Applications Containing Deficiencies, Errors, or Omissions
- (1) If the Utility identifies that the initial Application is not complete and contains deficiencies, errors, or omissions (collectively, "defects"), the Utility shall identify all such defects and shall return the Application to the Applicant within the ten (10) business days as indicated in Section E.2.g, specifying all such defects in sufficient detail to enable the Applicant to correct the Application for resubmission.
- (a) For Net Metering Systems, such identification of defects shall additionally include adherence to the technical requirements of NRS 704.774.
- (2) Any Application containing defects that are not fully corrected by the Applicant after two subsequent Utility reviews of the Application in addition to the Utility's first review may be rejected at the Utility's discretion, provided the Utility and the Applicant have worked in good faith, and in a timely manner, to assist the Applicant in submitting a correct and complete Application.

Commented [ZW8]: VGIC recommends V2G systems be eligible for the \$189 technical processing fee applied to Net Metering Systems. Note the residential V2G systems in particular are similar in nameplate export capability and, in some cases, use the very same inverter models used in NEM systems.

We understand this would be addressed in NVE's GRC, not in this Rule 15 revision. Because it is unlikely VGIC will have the resources to engage in NVE's next GRC, we respectfully request NVE consider proposing this modification in its next GRC.

- (3) An Application that, in the Utility's reasonable judgment, is materially incomplete, lacking substantial required information, or submitted solely to obtain position without a good-faith effort to provide a correct and complete submission, shall be rejected. In such cases, the Applicant must submit a new Application with all applicable fees. The new Application will be treated as an initial submission.
- (4) Any defect(s) present in an Application that is(are) identified by the Utility in one of its subsequent reviews of the Application following its initial review that the Utility failed to identify in any of its previous reviews of the Application shall reset the iterative review process begun in Section E.2.h.(1).
- (5) Any additional defects in a subsequent submission of an Application identified by the Utility that were not present in the initial submitted Application and which occurred in one of the subsequent submissions of the Application shall not reset the iterative review process begun in Section E.2.h.(1).
- (6) If the Utility rejects an Application in accordance with and following completion of the iterative review process described above, the Applicant shall submit a new Application and remit a new Application fee if the Applicant wishes to continue the application process. The Utility shall review the new Application for completeness and correctness as if it is an initial submission of the Application in accordance with Section E.2.g.
- (7) An Applicant may dispute the Utility's adherence to the above iterative review process and correctness in identifying defects in its initial or subsequently submitted Applications in accordance with Section K.
- i. The Utility shall record the date and time it accepts the Application as complete and correct and inform the Applicant of such date in a timely manner.

3. **Modifications to Applications or Existing Generating Facilities**

- a. The Applicant or the Producer shall notify the Utility of a modification to an Application or the equipment or operation of an existing Generating Facility in a timely manner or as required by the Interconnection and Operating Agreement.
- b. When notifying the Utility of a modification to an Application or the equipment or operation of an existing Generating Facility, the Applicant or the Producer shall include the relevant Generating Facility information (e.g., system rating, Advanced Inverter/PCS specifications, POI, as applicable) either originally approved or within the Application, together with the proposed changes and the elements to be modified clearly identified.
- c. Modifications to an accepted Application before execution of the Interconnection and Operating Agreement or issuance of PTO, and modifications to the equipment or operation of an existing Generating Facility, shall be reviewed under Table 2 to determine whether they constitute a Material Modification and the corresponding actions required of the Applicant or Producer. Any modification that does not reasonably fall within the descriptions in Table 2 shall be addressed on a case-by-case basis. The Applicant or the Producer and the Utility shall make good faith efforts to determine if a modification constitutes a Material Modification and agree upon the resulting course of action in a timely manner, at the Utility's discretion.
- d. If it is determined by the Utility, either at its discretion or through the application of Table 2, that a new Application is required as a result of a Material Modification, the applicable provisions of Section E.2 and the technical review process in Section F shall apply.

Commented [ZW9]: Per 10/30 stakeholder discussion, VGIC recommends NV Energy clarify that adding a non-exporting bidirectional EV charging systems to an existing NEM system does not constitute a material modification requiring a new application. While we understand not every use case can be accounted for, we do anticipate this being a common use case, given the propensity of rooftop solar customers to be aware of and interested in bidirectional charging systems.

TABLE 2

Section	Modification to Applications or Existing Generating Facilities	Notification Required?	New Application required?	Proceed without Utility approval?	Updated IOA required?
	Like-for-like replacement of existing equipment that does not constitute Material Modification, with no increase in system output (vs. original IOA) and no change to generator operating modes.	No	No	Yes	No
	Modifications where system capacity or output increases at the PCC and no inverter power or PCS controls are used to limit the active power to the original IOA size, including:	Yes	Yes	No	Yes
	Replacement of existing equipment	Yes	Yes	No	Yes
	Inverter firmware upgrade materially affecting grid interaction	Yes	Yes	No	Yes
	Change(s) to inverter characteristics that affect grid interaction	Yes	Yes	No	Yes
	Modifications the operation of the Generator(s)	Yes	Yes	No	Yes
	Change to a different POI or PCC than stated in the current IOA	Yes	Yes	No	Yes
	Sole change is ownership or Producer	Yes	No	No	Yes

- e. An Applicant or a Producer may dispute the Utility's determination of whether a modification constitutes a Material Modification or the actions required as a result of a modification in accordance with Section K, subject to and in accordance with the Interconnection and Operating Agreement.

F. **TECHNICAL REVIEW PROCESS**

1. Technical Review Level Determination and Process
 - a. Upon the Utility's acceptance of a satisfactorily-completed Application, including proposed Net Metering Systems, any additional information necessary to evaluate the Interconnection of a Generating Facility, and receipt by the Utility of any applicable processing fee, the Utility shall begin performing technical review. The business day response times indicated below that the Utility shall meet for review of the RPA, Simplified Review, Fast Track Review, and Supplemental Review begin upon the Utility's receipt of all the applicable information and fees in this Section.
 - b. The Utility shall determine which level of technical review is required as follows:
 - (1) If the proposed Generating Facility is a Certified inverter-based resource and has a Net Nameplate Rating of 50 kW or less and a Maximum Generating Capacity of 25-30 kW or less, then the Application qualifies to begin the technical review process with Simplified Review. Otherwise, the technical review process for the Application shall begin with either Fast Track Review or Supplemental Review.
 - (2) If the Application does not qualify for Simplified Review, based upon the proposed POI for the Generating Facility and its Maximum Generating Capacity, the Utility shall apply Table 3 below to determine whether the Application may begin the technical review process with Fast Track Review or Supplemental Review.

Commented [ZW10]: Based on conversations with VGIC members, we respectfully request that this cut-off be increased to 30 kW to better align with anticipated bidirectional charging system offerings.

TABLE 3

LINE VOLTAGE	FAST TRACK ELIGIBILITY	
	Regardless of location	On $\geq$ 600-amp line and $<$ 2.5 miles from substation
< 5 kV	< 1 MW	< 2 MW
5 kV – < 15 kV	< 2 MW	< 3 MW
15 kV – < 30 kV	< 3 MW	< 4 MW
30 kV – 69 kV	≤ 4 MW	≤ 5 MW

- (3) If the Maximum Generating Capacity of the proposed Generating Facility meets the criteria in Table 3 based upon the line voltage at the proposed POI, then the Application qualifies to begin the technical review process with Fast Track Review. Otherwise, the technical review process for the Application shall begin with Supplemental Review.
- c. At any time during the Application or technical review process, an Applicant may request the Utility to proceed to Detailed Review in lieu of performing Simplified, Fast Track, or Supplemental Review.
- d. The Utility shall inform the Applicant in a timely manner after its determination of the level at which the technical review process will begin.
- e. Confirmation of the RPA
- (1) The Applicant shall provide a proposed RPA on the Application. Unless otherwise approved, the default RPA shall be the PCC.
- (2) The Utility shall complete its review of the proposed RPA within five (5) business days, shall determine if it is appropriate, and shall inform the Applicant of such determination in a timely manner.
- (3) If the Utility determines the proposed RPA to be inappropriate, the Utility and the Applicant shall make good faith efforts to expeditiously meet and agree upon the RPA so as to not appreciably delay the technical review process.
- f. The Utility shall conduct its technical review based on the RPA, where applicable. The RPA shall be contained in the Net Metering System agreement or the Interconnection and Operating Agreement.
- g. The Utility shall perform its technical review process in accordance with the flowchart in Attachment 1.
- h. The Utility shall perform its technical review and determine the Interconnection requirements for each phase of a multi-phased Generator or Generating Facility Application at a single PCC (or RPA, if not the PCC) with consideration of previous phases but not future planned phases. Based on the requirements of future phases, amendments to existing agreements, including Interconnection and Operating Agreements, may be required, or existing agreements may be terminated by the Utility or the Producer and new agreements may be required.

2. Simplified Review

- a. The Utility shall complete its Simplified Review, absent any extraordinary circumstances, within ten (10) business days if the Application qualifies for Simplified Review and shall notify the Applicant if the Generating Facility meets all the applicable Simplified Process screens in a timely manner.
- b. The Utility's Simplified Review shall be performed in accordance with the flowchart in Attachment 1 and shall consist of the following Screens:

- (1) Screen A: Networked Secondary System
 - (a) The Utility shall determine whether the proposed Generating Facility would Interconnect to a networked secondary system.
 - (b) If the proposed Generating Facility would not Interconnect to a networked secondary system, then the Application passes this Screen. Otherwise, if it would, the Application fails this Screen.
- (2) Screen B: Certified Equipment
 - (a) On the basis of the information provided in the Application or any additional information provided to the Utility by the Applicant or the Producer either at the Utility's request or subsequent to the original Application, the Utility shall determine to the best of its ability if all of the equipment proposed for the Generating Facility is Certified Equipment per the requirements set forth in this Rule or if it will temporarily approve certain equipment that is not certified.
 - (b) If all the equipment proposed for the Generating Facility is Certified Equipment or if the Utility has granted, at its discretion on a case-by-case basis, interim approval of certain equipment that is not certified, then the Application passes this Screen. Otherwise, if all the equipment is not Certified Equipment and does not have interim approval by the Utility, then the Application fails this Screen.
- (3) Screen C: Transformer and Conductor Rating
 - (a) This Screen is not applicable to Interconnections on the primary side of a service transformer. Such Interconnections automatically pass this Screen.
 - (b) If the proposed Generating Facility is to Interconnect on the secondary side of a service transformer, the Utility shall determine if the addition of the Generating Facility would cause the normal continuous rating of either the service transformer or secondary conductor to which the Generating Facility would be Interconnected to be exceeded.
 - (c) For a Generating Facility proposed to Interconnect to a service transformer, if the addition of the proposed Generating Facility, when considering the total aggregate Maximum Generating Capacity of all existing Generating Facilities Interconnected to the service transformer or secondary conductor and that of the proposed Generating Facility, would not exceed the normal continuous ratings of both the service transformer and secondary conductor, then the Application passes this Screen. Otherwise, if it would exceed either the normal continuous ratings of the service transformer or secondary conductor, or both, then the Application fails this Screen.
- (4) Screen D: Service Imbalance
 - (a) If the proposed Generating Facility is single-phase and is to Interconnect on a center tap neutral of a 240 volt service, the Utility shall determine whether the addition of the Generating Facility would cause excessive imbalance between the two sides of the 240 volt service.
 - (b) If the addition of the proposed Generating Facility would create an imbalance of twenty (20) percent or less of the normal continuous nameplate rating of the service transformer, then the Application passes this Screen. Otherwise, if it would create an imbalance of more than twenty (20) percent of the normal continuous nameplate rating of the service transformer, then the Application fails this Screen.

(5) Screen E: Hosting Capacity or Line Section Load

- (a) If Hosting Capacity values are available at the Generating Facility's POI, the Utility shall compare those values to the Generating Facility's Maximum Generating Capacity.
 - i. If the Maximum Generating Capacity of the Generating Facility and/or the load imposed on the Utility's Distribution System by the Generating Facility (e.g., for a ESD) is less than or equal to ninety (90) percent of the appropriate Hosting Capacity value, then the Application passes this Screen. Otherwise, if it is greater than ninety (90) percent of the appropriate Hosting Capacity value, then the Application fails this part of this Screen.
 - ii. If the Application fails this part of this Screen, then the Utility shall apply the next part of this Screen (Section F.2.b.(5)(b) – Relevant Minimum Load). If the Application passes that part of this Screen, then the Application passes this Screen despite failing the Hosting Capacity part of this Screen.
- (b) If Hosting Capacity values are not available and cannot be reasonably estimated or calculated at the Generating Facility's POI or if Section F.2.b.(5)(a)(i) has been failed, and where twelve (12) months of Line Section minimum load data (including onsite load but not station service load served by the proposed Generating Facility) are available or can be estimated or calculated from existing data or from a power flow model, the Utility shall 1. determine the Relevant Minimum Load and 2) compare that value to the total aggregate Generating Facility Maximum Generating Capacity on the Line Section (existing plus the proposed Generating Facility) that the Generating Facility will Interconnect to and to all Line Sections bounded by automatic sectionalizing devices upstream of the proposed Generating Facility. If adequate load data is not available and the Utility is unable to reasonably estimate or calculate the Line Section Relevant Minimum load, the Utility shall include the reason(s) that it is unable to do so in its technical review results.
 - i. If the Utility's comparison yields a one hundred (100) percent or less result(s), then the Application passes this Screen. Otherwise, if it yields a greater than a one hundred (100) percent result(s), then the Application fails this Screen.
- (c) If both Hosting Capacity and Relevant Minimum Load values are not available and cannot be reasonably estimated or calculated at the Generating Facility's POI, the Utility shall compare the total aggregate Generating Facility Maximum Generating Capacity on the Line Section (existing plus the proposed Generating Facility) to fifteen (15) percent of the peak (maximum) load of the Line Section that the Generating Facility will Interconnect to. If adequate load data is not available and the Utility is unable to reasonably estimate or calculate the Line Section peak load, the Utility shall include the reason(s) that it is unable to do so in its technical review results.
 - i. If the Utility's comparison yields a fifteen (15) percent or less result, then the Application passes this Screen. Otherwise, if it yields a greater than fifteen (15) percent result, then the Application fails this Screen. However, if the Generating Facility has been designated as Non-Exporting, then it automatically passes this part of this Screen.
- c. If any of the Simplified Review Screens are failed, the Utility at its discretion may approve the Interconnection provided such approval shall not impact the safety and

reliability of its Distribution System. Otherwise, if the Utility does not approve the Interconnection, the Application shall proceed to the Applicant's options process pursuant to Section F.5.

- d. If all Simplified Review Screens are passed, then the proposed Generating Facility may be Interconnected without further technical review. The Utility shall provide the Applicant with a written description of the requirements for Interconnection and the Interconnection and Operating Agreement pursuant to Section E.2. in a timely manner if it has not already provided such agreement to the Applicant.

3. Fast Track Review

- a. The Utility shall begin Fast Track Review upon its determination that this review level is required to determine the Interconnection requirements for the Application based upon its technical review level determination, unless the Applicant requests that the Utility perform either Supplemental or Detailed Review in lieu of performing Fast Track Review. If the Utility determines that additional information is required to perform Fast Track Review, the Applicant and the Utility shall make good faith efforts to request and provide such information in a timely manner.
- b. The Utility shall complete its Fast Track Review, absent any extraordinary circumstances, within fifteen (15) business days if the Application qualifies for Fast Track Review and shall notify the Applicant if the Generating Facility meets all the applicable Fast Track Review Screens in a timely manner.
- c. The Utility's Fast Track Review shall be performed in accordance with the flowchart in Attachment 1 and shall include the Simplified Review Screens A through E in addition to the following Screens F through J below:

(1) Screen F: Short Circuit Current Contribution

- (a) Inverter-based Generating Facilities with a Net Nameplate Rating of less than ~~25-30~~ kW automatically pass this Screen.
- (b) The Utility shall determine if the addition of the Generating Facility would cause allowable Short Circuit Contribution Ratio limits to be exceeded by ensuring that both of the following criteria are met (when applicable):
 - i. When measured at primary side (high side) of a Dedicated Distribution Transformer serving a Generating Facility, the sum of the Short Circuit Contribution Ratios of all Generating Facilities connected to the particular Distribution System feeder that serves the Generating Facility must be less than or equal to 0.1; and
 - ii. When measured at the secondary side (low side) of a shared distribution service transformer, the short circuit contribution of the proposed Generating Facility must be less than or equal to 2.5 percent of the interrupting rating of the Producer's service equipment.
- (c) If the addition of the proposed Generating Facility, when considering the total aggregate Net Nameplate Ratings of all existing Generating Facilities Interconnected to the particular Distribution System feeder and the proposed Generating Facility meets both of the above criteria, then the Application passes this Screen. Otherwise, if it does not meet one or both of the above criteria, then the Application fails this Screen.

(2) Screen G: Short Circuit Interrupting Capability

- (a) The Utility shall determine if the addition of the Generating Facility would cause the allowable Short Circuit Interrupting Capability of the Utility's protective devices or equipment on the Distribution System feeder or the Producer's protective devices or equipment to be exceeded.

Commented [ZW11]: See VGIC comment in Section F.1.b.(1)

- (b) If the addition of the proposed Generating Facility, when considered in aggregate with the Net Nameplate Ratings of all existing Generating Facilities on the Distribution System feeder, would not cause the Utility's protective devices or equipment on the Distribution System feeder or the Producer's protective devices or equipment to exceed 87.5 percent of their Short Circuit Interrupting Capability, then the Application passes this Screen. Otherwise, if it would cause the Utility's protective devices or equipment on the Distribution System feeder or the Producer's protective devices or equipment to exceed 87.5 percent of their Short Circuit Interrupting Capability, then the Application fails this Screen.

(3) Screen H: Line Configuration

- (a) Single-phase inverter-based Generating Facilities automatically pass this Screen.
- (b) The Utility shall determine if the Interconnection of the proposed Generating Facility passes this Screen in accordance with the following:
 - i. If the primary Distribution System line configuration is three-phase, three-wire, then this Screen is passed for any type of Interconnection to the primary Distribution System line; or
 - ii. If the primary Distribution System line configuration is three-phase, four wire, then this Screen is passed if the Interconnection to the primary Distribution System line is single-phase, line-to-neutral; or
 - iii. If the primary Distribution System line configuration is three-phase, three-wire (for any line that has such a section or mixed three-wire and four-wire), then this Screen is passed for all other types of Interconnections to the primary Distribution System line.
 - iv. Otherwise, this Screen is failed.

(4) Screen I: Inadvertent Export

- (a) This Screen only applies when the Net Nameplate Capacity of the Generating Facility minus its Maximum Generating Capacity is greater than 250 kW for a Generating Facility that can introduce Inadvertent Export. Otherwise, if that difference is 250 kW or less, this Screen is automatically passed.
- (b) The Utility shall determine if the change in voltage at the point on the Utility's primary Distribution System nearest the POI due to Inadvertent Export of the Generating Facility does not exceed three (3) percent. The voltage change shall be estimated by applying the formula:

$$\frac{(R_{\text{source}} \times \Delta P) - (X_{\text{source}} \times \Delta Q)}{V^2}, \text{ where:}$$

- i. R_{source} equals the grid resistance;
- ii. ΔP equals (Net Nameplate Rating - the Maximum Generating Capacity) x power factor;
- iii. X_{source} equals the grid reactance;
- iv. ΔQ equals (Net Nameplate Rating - the Maximum Generating Capacity) x $\sqrt{1 - \text{power factor}^2}$; and
- v. V equals the primary Distribution System voltage.

- (c) If the result of the above formula is three (3) percent or less, then the Application passes this Screen. Otherwise, if it is greater than three (3) percent, then the Application fails this Screen.

(5) Screen J: Starting Voltage Dip

- (a) This Screen only applies to Generating Facilities that start by motoring the Generator(s). Inverter-based Generators automatically pass this Screen.
 - (b) The Utility shall determine if the starting of the Generator(s) associated with a proposed Generating Facility would exceed the maximum allowable voltage dip on its Distribution System by using either of the following two options at its discretion:
 - i. Determining that the Generating Facility's starting In-rush Current is equal to or less than the continuous ampere rating of the Producer's service equipment; or
 - ii. Determining the impedances of the distribution service transformer (if present) and the secondary conductors to the Producer's service equipment and performing a voltage dip calculation. Alternatively, the Utility may use tables or nomographs to determine the voltage dip. Voltage dips caused by starting a Generator as a motor must be less than two and one-half (2.5) percent for primary Distribution System Interconnections and five (5) percent for secondary Distribution System Interconnections.
 - (c) If, by employing either of the above two options, the Utility determines that the starting voltage dip of the Generator(s) associated with a proposed Generating Facility would not exceed the maximum allowable voltage dip on its Distribution System, then the Application passes this Screen. Otherwise, if it would, then the Application fails this Screen.
 - (d) Passing this Screen still requires the Producer to ensure that its Generating Facility complies with the flicker requirements of Section G.3.f.
- d. If any of the Fast Track Review Screens are failed, the Utility at its discretion may approve the Interconnection provided such approval shall not impact the safety and reliability of its Distribution System. Otherwise, the Utility shall begin Supplemental Review unless the Applicant requests an Applicant's options meeting pursuant to Section F.5 in lieu of the Utility performing Supplemental Review.
- e. If all Screens A through J are passed, the proposed Generating Facility may be Interconnected without further technical review. The Utility shall provide the Applicant with a written description of the requirements for Interconnection and the Interconnection and Operating Agreement pursuant to Section E.2 in a timely manner if it has not already provided such agreement to the Applicant.

4. Supplemental Review

- a. The Utility shall begin Supplemental Review upon its determination that this review level is required to determine the Interconnection requirements for the Application based upon its technical review level determination or if the Application has failed Fast Track review, unless the Applicant requests that the Utility perform Detailed Review in lieu of performing Supplemental Review. If the Utility determines that additional information is required to perform Supplemental Review, the Applicant and the Utility shall make good faith efforts to request and provide such information in a timely manner.
- b. The Utility shall complete its Supplemental Review, absent any extraordinary circumstances, within twenty (20) business days if the Application qualifies for

Supplemental Review and shall notify the Applicant if the Generating Facility meets all the applicable Supplemental Review Screens in a timely manner.

- c. The Utility's Supplemental Review shall be performed in accordance with the flowchart in Attachment 1 and shall consist of Screens A through J (if not already performed as part of Fast Track Review) in addition to the following Screens K through M below:

(1) Screen K: Supplemental Grounding

- (a) If Screen H was failed, then the Utility shall apply this Screen. If Screen H was passed, then this Screen is automatically passed.
- (b) If the Generator is rotating machine-based, then effective grounding must be maintained for this Screen to be passed. Otherwise, this Screen is failed.
- (c) If the Generator is inverter-based, then the Utility shall apply one of the following to evaluate if it is effectively grounded:
 - i. If the line-to-neutral connected load on the Distribution System feeder or Line Section is greater than thirty-three (33) percent of the peak load on the Distribution System feeder or Line Section, then this Screen is passed; or
 - ii. If a supplemental grounding software tool is used and it determines that supplemental grounding is not required to maintain effective grounding, then this Screen is passed; or
 - iii. If a supplemental grounding software tool is used and it determines that supplemental grounding is required, and the Applicant agrees to modify its system to include supplemental grounding, then this Screen is passed; or
 - iv. If a detailed Hosting Capacity analysis is used that incorporates evaluation of temporary overvoltage risk for inverters and if the Net Nameplate Rating of the Generator is below the available Hosting Capacity at the POI, then this Screen is passed.
 - v. Otherwise, if none of the above four criteria are met, then the Application fails this Screen.

(2) Screen L: Voltage and Power Quality

- (a) The Utility shall determine if the addition of the Generating Facility would cause voltage fluctuation or harmonic levels on the Line Section to be exceeded.
- (b) Considering the total aggregate Generating Facility Maximum Generating Capacity on the Line Section (existing plus the proposed Generating Facility):
 - i. The voltage fluctuation shall be within acceptable limits as defined by IEEE 1547-2018; and
 - ii. The harmonic levels shall meet IEEE 1547-2018 limits at the RPA.
- (c) If both of the above two criteria are met, then the Application passes this Screen. Otherwise, if either or both of the above two criteria are not met, then the Application fails this Screen.

(3) Screen M: Safety and Reliability

- (a) The Utility shall consider that following factors to determine if the Generating Facility may potentially impact safety and reliability on the Utility's Distribution System:

- i. If the Line Section has significant minimum loading levels dominated by a small number of customers (i.e., several large commercial customers);
- ii. If the loading along the Line Section is uniform or evenly distributed;
- iii. If the proposed Generating Facility is located in close proximity to the substation (i.e., less than two and one-half (2.5) electrical circuit miles), and if the Line Section from the substation to the PCC is a main backbone section of the Distribution System feeder rated for normal and emergency ampacity;
- iv. If the proposed Generating Facility incorporates a time delay function to prevent reconnection of the Generator(s) to the electrical Distribution System until system voltage and frequency are within normal limits for a prescribed time;
- v. If operational flexibility is reduced by the proposed Generating Facility, such that transfer of the Line Section(s) to which the Generating Facility is Interconnected to a neighboring distribution feeder/substation may trigger overloads or voltage issues; and
- vi. If the proposed Generating Facility employs equipment or systems Certified by a recognized standards organization or NRTL to address technical issues such as, but not limited to, islanding, reverse power flow, or voltage quality.

(b) If all of the above six conditions are met, then the Application passes this Screen. Otherwise, if any of the above six conditions are not met, then the Application fails this Screen.

- d. If any of Screens A through M are failed, the Utility at its discretion may approve the Interconnection provided such approval shall not impact the safety and reliability of its Distribution System. Otherwise, the Application shall proceed to the Applicant's options process pursuant to Section F.5.
- e. If all Screens A through M are passed, the proposed Generating Facility may be Interconnected without further technical review. The Utility shall provide the Applicant with a written description of the requirements for Interconnection and the Interconnection and Operating Agreement pursuant to Section E.2. in a timely manner if it has not already provided such agreement to the Applicant.

5. Applicant's Options

- a. When the Utility determines that the Application does not pass Simplified Review, Fast Track Review, or Supplemental Review, it shall notify the Applicant of such determination and inform the Applicant which Screen(s) were failed and why in a timely manner. In this instance, the Utility and the Applicant shall make good faith efforts to expeditiously schedule and hold a meeting to discuss the Applicant's options. Additionally, the Applicant may request that the Utility schedule an Applicant options meeting at any time during the technical review process.
- b. Among any other option mutually agreed upon between the Utility and the Applicant, the Applicant shall have the following options:
 - (1) Withdraw the Application;
 - (2) Amend the Application and request the Utility to re-perform the specific failed technical review Screen(s); or
 - (3) Request the Utility to begin the Detailed Review process.

- c. If the Applicant chooses to amend the Application to address the specific failed technical review Screen(s):
 - (1) the Applicant shall submit to the Utility an amended Application demonstrating the redesign within ten (10) business days after the Applicant options meeting. If the amended Application is not received by the Utility within ten (10) business days, the Utility shall consider the Application withdrawn;
 - (2) an amended Application shall only include changes to address the specific failed technical review Screen(s) and only those other changes agreed to by the Utility at its discretion. Any other changes to the Application shall be addressed in accordance with Section E.3; and
 - (3) the Utility shall review the amended Application, apply the failed technical review Screen(s), determine if the amended Application addresses the technical review Screen(s) failure(s), and shall notify the Applicant of the results of this review within ten (10) business days of receiving the amended Application.
- d. The amended Application option to mitigate the failure of a technical review Screen(s) shall only be available once during the technical review process. Subsequent amending of the Application shall result in the Utility considering it as a new Application and reinitiating the technical review process from its beginning in accordance with Section E.3. At its discretion, the Utility may also require reinitiation of the technical review process of an Application if an Applicant amends information in an Application that has already completed the technical review process, including but not limited to, on the basis of changes in the size(s), the proposed POI, PCC or RPA, anticipated commercial operation date(s), or planned operation or operating mode(s) of the Generating Facility.
- e. If the Applicant chooses to request the Utility begin Detailed Review, the Applicant shall submit to the Utility such request within ten (10) business days after it is notified by the Utility that the Application has been determined to not pass Simplified Review, Fast Track Review, or Supplemental Review, or the Applicant's options meeting if one was conducted. If such notification is not received by the Utility within ten (10) business days, the Utility shall consider the Application withdrawn.

6. Detailed Review

- a. When the Utility's technical review process reveals that the proposed Generating Facility Application does not pass all applicable technical Screens and the Applicant is either unable to amend the Application adequately or if the Applicant requests the Utility to begin Detailed Review at any time during the technical review process, the Utility shall conduct an Interconnection Study to determine if any significant Utility Interconnection Facilities or Distribution System improvements must be installed or be made to the Utility's Distribution System to accommodate the Interconnection of an Applicant's Generating Facility.
- b. The Utility and the Applicant shall enter into an Interconnection Study Agreement that provides for the Utility to perform an Interconnection Study, facility design and engineering, and to provide detailed cost estimates for actual cost billing to the Applicant at the Applicant's expense. The Interconnection Study Agreement shall set forth the Utility's schedule for completing such work and the estimated or fixed price costs of such studies and engineering.
- c. Upon completion of an Interconnection Study, the Utility shall provide the Applicant with the specific requirements, details of violations of any design, planning, or operating criteria or standards including estimated magnitude, duration, and/or frequency, as applicable, estimated costs, and schedule for Interconnecting the Generating Facility to accommodate execution of agreements pursuant to Section E.2. in a timely manner.
- d. Upon receipt of any required Interconnection Facilities or Distribution System improvements from the Utility as a result of the Interconnection Study, within ten (10) business days the Producer shall:

- (1) determine if the Generating Facility's operating characteristics can be modified such that any required Interconnection Facilities or Distribution System improvements could be reduced or not required (e.g., through the use of a PCS or Limited Generation Profile operation); and
 - (2) inform the Utility of such capability and request the Utility to determine the need for and scope of any Interconnection Facilities or Distribution System improvements based upon the operating characteristics that the Producer indicates can be modified; or
 - (3) inform the Utility that no such modifications to the Generating Facility's operating characteristics can be made that would reduce or preclude the need for the required Interconnection Facilities or Distribution System improvements.
- e. Upon receipt of a request from the Applicant, the Utility shall then determine and inform the Producer of what modifications, if any, may be made to the required Distribution System improvements based upon the Producer's modified operating characteristics, or shall inform the Producer of the time frame to make such a determination, within ten (10) business days of receipt of such modified operating characteristics. Modifications to an initial Application that do not require submission of a new Application shall not incur an additional study fee.

G. DESIGN, TECHNICAL AND OPERATING REQUIREMENTS

1. General Interconnection Requirements
 - a. Where there is a conflict between the technical requirements of IEEE 1547-2018 and this Rule, the technical requirements of IEEE 1547-2018 apply.
 - b. Enter Service requirements shall conform to IEEE 1547-2018 Table 4 with the intentional adjustable minimum delay, the Enter Service period (ramp duration), and the randomized time delay maximum interval each set to 300 seconds as a default. For Generators with a Net Nameplate Rating of less than 500 kVA, individual Generators may use the randomized time delay as an alternative to the Enter Service period (ramp duration).
 - c. The protective functions and requirements of this Rule are designed to protect the Utility's electrical Distribution System and not the Generating Facility. A Producer shall be solely responsible for providing adequate protection for its Generating Facility and Interconnection Facilities. The Producer's protective devices shall not impact the operation of other protective devices used on the Utility's Distribution System in a manner that would affect the Utility's capability to provide reliable service to its Customers.
 - d. Prior to Parallel Operation or Momentary Parallel Operation of the Generating Facility, the Utility shall approve the Producer's Protective Function and control diagrams. Generating Facilities equipped with a Protective Function and control scheme previously approved by the Utility for system-wide application or only Certified Equipment may satisfy this requirement by reference to previously approved drawings and diagrams.
 - e. In the event that this Section of the Rule does not address the Interconnection requirements for a particular Generating Facility, the Utility shall specify the technical requirements to Interconnect the Generating Facility.
2. Non-Exporting and Limited Export Generating Facilities
 - a. Inadvertent Export from a Generating Facility shall comply with the limits identified in this Section for a Generating Facility to be designated as Non-Exporting or Limited Export.
 - b. The Generating Facility is connected on the load-side of the Customer's main breaker.

- c. A Generating Facility shall incorporate one of the following options to be designated as Non-Exporting:
- (1) Option 1 ("Reverse Power Protection"): To ensure power is never Exported, a reverse power Protective Function shall be implemented at the PCC. The default setting for this Protective Function, when used, shall be 0.1 percent (Export) of the service transformer's rating, with a maximum two (2.0) second time delay.
 - (2) Option 2 ("Minimum Power Protection"): To ensure at least a minimum amount of power is imported at all times (and, therefore, that power is not Exported), an under-power Protective Function shall be implemented at the PCC. The default setting for this Protective Function, when used, shall be five (5) percent (import) of the Generating Facility's total Gross Nameplate Rating, with a maximum two (2.0) second time delay.
 - (3) Option 3 ("Certified Non-Islanding Protection"): To ensure that Inadvertent Export is limited to acceptable levels, this option, when used, requires that all the following conditions be met:
 - (a) The total Gross Nameplate Rating of the Generating Facility shall be no more than 25 percent of the nominal ampere rating of the Producer's service equipment;
 - (b) The total Gross Nameplate Rating of the Generating Facility shall be no more than 50 percent of the Producer's service transformer capacity rating (this capacity requirement does not apply to Customers taking primary service without an intervening transformer); and
 - (c) The Generating Facility shall be certified as Non-Islanding.
 - (4) Option 4 ("Relative Unit Size"): This option, when used, requires the Net Nameplate Rating of the Generating Facility to be so small in comparison to its host facility's minimum Host Load, that the use of additional Protective Functions is not required to ensure that power will not be Exported to the Utility's Distribution System. The Generating Facility's Net Nameplate Rating shall be no greater than 50 percent of the Producer's verifiable minimum Host Load over the past 12 months.
- d. A Generating Facility shall incorporate one of the following options to be designated as Limited Export:
- (1) Option 5 ("Directional Power Protection"): This option, when used, requires the use of a power Protective Function at the PCC. The default setting for this Protective Function shall be the Maximum Generating Capacity value with a maximum two (2.0) second time delay.
 - (2) Option 6 ("Configured Power Rating"): A reduced output rating utilizing the power rating configuration setting of a Generator (if available) may be used to ensure that the Generator or Generating Facility does not generate electric power beyond a certain value that may be lower than the Gross Nameplate Rating or Net Nameplate Rating. The configuration setting corresponds to the active or apparent power ratings in Table 28 of IEEE Std 1547-2018, as described in subclause 10.4. A local DER communication interface is not required to utilize the configuration setting if it can be set by other Certified means. The reduced power rating may be indicated by means of a Gross Nameplate Rating or Net Nameplate Rating replacement, or a supplemental adhesive de-rating tag to indicate the reduced power rating. This option, when used, shall be Certified to IEEE 1547.1-2020. Use of a configured power rating not applied to individual Generator(s) shall require evaluation by mutual agreement between the Utility and the Producer.

- e. A Generating Facility may be designated as either Non-Exporting or Limited Export if it is designed with a Certified PCS, and if inverter-based, Certified Advanced Inverter(s), to enforce levels of Limited Export and Inadvertent Export by mutual agreement between the Utility and the Producer (this is Option 7). This option, when used, shall have a maximum open loop response time of no more than 30 seconds to ensure that Inadvertent Export is limited to acceptable levels.
- f. A Generating Facility may be designated as either Non-Exporting or Limited Export if it is designed with other control systems and/or Protective Functions to limit Export and Inadvertent Export by mutual agreement between the Utility and the Producer (this is Option 8). The limits may be based on technical limitations of the Interconnection Customer's equipment or the Utility's Distribution System equipment. To ensure that Limited Export and Inadvertent Export levels remain within mutually agreed-upon limits, the Interconnection Customer may use an uncertified PCS, an internal transfer relay, energy management system, or other Customer facility hardware or software if approved by the Utility on a case-by-case basis at its discretion.

3. Protective Functions and Safety Equipment Requirements

- a. The Protective Functions for Generating Facilities operating in Parallel Operation with the Utility's Distribution System shall consist of an effective means of quickly detecting and protecting against the formation of an Unintentional Island on the Utility's Distribution System. Such Protective Functions shall trip and disconnect the Generating Facilities from the Utility's Distribution System and shall be able to sense and prevent reconnection when the Utility's Distribution System is de-energized. Protective Functions shall be subject to Utility approval and may include, but are not limited to:
 - (1) Definite-time and/or inverse-time under-/over-voltage element(s);
 - (2) Definite-time under-/over-frequency element(s);
 - (3) Rate-of-change of frequency element(s);
 - (4) Directional power and/or over-/under-power element(s);
 - (5) Transfer Trip Capability (meaning to receive a direct transfer trip from the Utility's electrical system protective devices); and
 - (6) Disconnecting Means Status (meaning to communicate the Generating Facility's Disconnecting Means status to the Utility's electrical system protective devices using "b-form" contacts).
- b. Circuit breakers or other interrupting devices located at the PCC must be Certified or "Listed" (as defined in Article 100, the Definitions Section of the National Electrical Code) as suitable for their intended application. This includes being capable of interrupting the maximum available fault current expected at their location. The Producer's Generating Facility and Interconnection Facilities shall be designed so that the failure of any one device shall not potentially compromise the safety and reliability of Utility's Distribution System.
- c. The Producer shall furnish and install a manual Visible-Break Isolation Device that has a visible disconnect to isolate the Generating Facility from the Utility's Distribution System. The device must be accessible to Utility personnel at all times and be capable of being locked in the open position. Generating Facilities with Non-Islanding inverters totaling one (1) kVA or less are exempt from this requirement.

4. Additional Requirements to Preclude Interference with Utility Operations

- a. The Producer shall not operate Generating or Interconnection Facilities that superimpose a voltage or current upon the Utility's Distribution System that interferes with Utility operations, service to Utility Customers, or communication facilities. If such interference occurs, the Producer must diligently pursue and take corrective action at its own expense after being given notice and reasonable time to do so by the Utility. If the

Producer does not take corrective action in a timely manner or continues to operate the facilities causing interference without restriction or limit, the Utility may, without liability, disconnect the Producer's facilities from the Utility's Distribution System, in accordance with Section D.11 of this Rule. All operation shall conform to the power-quality performance requirements of IEEE 1547-2018 Clause 7 and the Utility's adopted default advanced inverter settings.

- b. Voltage interference causing a modulation of the light level of lamps sufficient to be irritating to humans, or which causes equipment mis-operation to occur shall be corrected by the Producer. Unacceptable interference shall be as defined in accordance with IEEE 1453-2022, as referenced by IEEE 1547-2018 Clause 7.2.3 and Annex B, or as otherwise specified by the Utility. Inverters certified to IEEE 1547-2018 and operated using the Utility's default advanced inverter settings shall be deemed compliant unless demonstrated otherwise.
- c. When the Producer is serving balanced linear loads, harmonic distortion shall be within the maximum harmonic current distortion percentages given in IEEE 1547-2018 Clause 7.3 and Table 11, evaluated at the Reference Point of Applicability (RPA). Exception: The harmonic distortion of a Generating Facility located at a Customer's site shall be evaluated using the same criteria as the loads at that site.
- d. Generating Facilities should not inject DC current greater than 0.5 percent of rated output current into the Utility's Distribution System, consistent with IEEE 1547-2018 Clause 7.4.
- e. Generating Facilities shall not contribute to an Unintentional Island. This can be accomplished by one of the following options:
 - (1) Incorporating certified Non-Islanding control functions into the Protective Functions; or
 - (2) Verifying that local loads sufficiently exceed the load carrying capability, including that the reactive power requirements of the local loads sufficiently exceed the reactive power capability of the Generating Facility; or
 - (3) Incorporating Transfer Trip or an equivalent function in the Protective Functions.

5. Certain Technology Specific Requirements

- a. For three-phase Generators, the Generating Facility circuit breakers shall be three-phase devices with electronic or electromechanical control. The Producer shall be responsible for properly synchronizing its Generating Facility with the Utility's Distribution System by means of either a manual or automatic synchronizing device, in accordance with the synchronization performance requirements of IEEE 1547-2018, Clause 4.1.3 and Table 5. Automatic synchronizing is required for all synchronous Generators that have a SCCR exceeding 0.05. A Generator whose SCCR exceeds 0.05 shall be equipped with Protective Functions suitable for detecting loss of synchronism and rapidly disconnecting the Generator from the Utility's Distribution System. As agreed upon by the Producer and the Utility, synchronous Generators may automatically regulate power factor or voltage while in Parallel Operation with the Utility's Distribution System. Power system stabilization is specifically not required for Generating Facilities under 10 MW Net Nameplate Rating. A Generating Facility with an SCCR exceeding 0.1 or that does not meet any one of the options for detecting Unintentional Islands in G.3.e. shall be equipped with Protective Functions designed to detect Distribution System faults, both line- to-line and line-to-ground, and promptly remove the Generating Facility from the Utility's Distribution System in the event of a fault. For a Generating Facility that cannot detect these faults within 0.5 seconds, Transfer Trip or an equivalent function may be required. Reclose-blocking of the Utility's affected recloser(s) may also be required by the Utility where engineering review indicates a risk of reclosing onto an energized island or where protective coordination cannot otherwise be assured.

- b. Induction Generators do not require a synchronizing function. Starting or rapid load fluctuations on induction generators can adversely impact the Utility's Distribution System's voltage. Corrective step-switched capacitors or other techniques may be necessary and may cause undesirable ferro-resonance. When these counter measures (e.g., additional capacitors) are installed on the Producer's side of the PCC, the Utility shall review these measures. Additional equipment may be required as determined as a result of Supplemental Review or an Interconnection Study.
- c. For inverter-based Generators, only Advanced Inverters shall be used for Parallel Operation with the Utility's Distribution System. Non-utility-interactive or "stand-alone" inverters shall not be used for Parallel Operation with the Utility's Distribution System.
- d. Generating Facilities with Advanced Inverters do not require separate synchronizing equipment.

6. Rotating Machine Generator Capabilities and Default Settings

- a. Rotating machine-based Generators shall meet IEEE 1547-2018 Category A for normal operating performance and Category I for abnormal operating performance. Rotating machine-based Generators that cannot meet these requirements shall be reviewed on a case-by-case basis by the Utility and authorized for Interconnection or not at its discretion.
- b. The default settings identified within this section shall be used by all rotating machine-based Generators regardless of size.
- c. Default Settings
 - (1) The Utility may specify alternate settings than those referenced in this section on a case-by-case basis and shall inform the Applicant and Producer, allowing reasonable time for mutual agreement of the parties prior to execution of the Interconnection and Operating Agreement.
 - (2) Volt-VAR Settings
 - (a) This setting shall be enabled. All other reactive power control modes shall be disabled.
 - (b) Rotating machine-based Generators shall utilize a fixed reference voltage for this setting.
 - (c) Default settings shall conform to IEEE 1547-2018 Table 8.
 - (3) Voltage Trip and Ride-Through
 - (a) Voltage trip settings shall conform to IEEE 1547-2018 Table 11.
 - (b) Voltage ride-through performance shall conform to IEEE 1547-2018 Table 14.
 - (4) Frequency Trip and Ride-Through
 - (a) Frequency trip settings shall conform to IEEE 1547-2018 Table 18.
 - (b) Frequency ride-through performance shall conform to IEEE 1547-2018 Table 19.
 - (c) Frequency droop performance shall conform to IEEE 1547-2018 Table 24.

7. Specific Energy Storage Device Requirements

- a. Energy Storage Device Operating Restrictions
 - (1) When an ESD is paired with a Net Metering System, the ESD, at the election of the Customer and/or Producer, may be programmed with either one or both of the following two operating restrictions, or no operating restriction, as follows:

- (a) Restricted from Exporting electricity to the Distribution System (beyond Inadvertent Export); and/or
 - (b) Restricted to being charged solely from the Producer's Net Metering System and not from the Utility's Distribution System; or
 - (c) No operating restrictions, only where expressly authorized under a Commission-approved tariff, Utility program, or Interconnection and Operating Agreement that governs the operating capabilities of the Energy Storage Device.
 - (2) If the Customer and/or Producer elects to operate an ESD with an operating restriction(s), such election shall be stated in the Application and Interconnection and Operating Agreement, including a description of the operating restriction(s).
 - (3) An attestation of the operating restriction(s), or lack of an operating restriction(s), shall be provided by both the Customer and the ESD Producer, if different entities.
 - (4) After the submission of the Application, the presence or absence of an operating restriction(s) may be modified, but not until after an updated attestation is provided to the Utility by both the Customer and the ESD Producer, if different entities, and the Utility consents in writing to the modification. The Utility's written approval shall not be unreasonably withheld or delayed.
- b. A stand-alone ESD Interconnected to the Utility's Distribution System that is not paired with a Generator, shall be charged solely from the Utility's Distribution System, and shall discharge only for the sole purpose of providing back-up service to all or part of a Customer's load during an interruption of electric service from the Utility shall first disconnect from the Utility's Distribution System before beginning to discharge within backup operation and shall cease to discharge prior to returning to normal operation Interconnected with the Utility's Distribution System.
- c. An ESD paired with a Generator Interconnected to the Utility's Distribution System that, in addition to its normal operation, shall discharge to provide back-up service to all or part of a Customer's load during an interruption of electric service from the Utility shall first disconnect from the Utility's Distribution System before beginning to discharge within backup operation and shall cease to discharge prior to returning to normal operation Interconnected with the Utility's Distribution System, after which it may discharge according to its normal operation.
8. Specific Electric Vehicle, Charger and Charging Station Requirements
- a. The Interconnection of a V2G EV and its EVSE to the Utility's Distribution System, or conversion of an EV charger, charging station, or EVSE to V2G, shall be subject to this Rule only when the V2G EV and its EVSE shall operate in a mode where discharging will occur when Interconnected and operating in Parallel Operation with the Utility's Distribution System.
 - b. V2G EV chargers or charging station inverters (i.e., DC bidirectional charging systems) shall adhere to UL 9741 or UL 1741 SB to ensure that V2G systems can safely export power from EVs to the Utility's Distribution System.
 - c. V2G EVs configured as an AC bidirectional charging system, wherein inverter capabilities are onboard the vehicle, with a built-in inverter shall also either adhere to standard SAE J3072 or be part of a UL 1741 CRD/UL 9741 matched pair DER system. EVSE used in these configurations should either adhere to UL 1741 SC (for SAE J3072 pathway) or be part of a UL 1741 CRD/UL 9741 matched pair DER system.
 - d. A V2G EV, EV charger, charging station, or EVSE connected to the Utility's Distribution System that shall discharge only for the sole purpose of providing back-up service to all or part of a Customer's load during an interruption of electric service from the Utility shall first disconnect from the Utility's Distribution System before beginning to discharge within

Commented [ZW12]: We respectfully urge NVE to adopt both pathways as options for customers/vendors to choose from. The market is relatively nascent, and limiting V2G DC interconnection to one pathway, despite both offering the core safety and grid interoperability requirements, would unnecessarily restrict the availability of these resources for customers.

Today's available product landscape already includes systems with integrated inverters in the charging station and standalone inverters outside of the charging station. For example, the current Ford and GM systems use UL 1741 SB certified inverters that are not housed within the charging wallbox.

Commented [ZW13R12]: Alternatively, this section could read: "V2G DC EVSE shall adhere to either:
(1) UL 9741
(2) UL 1741 SB"

This version would be very clear. It would also be consistent with interconnection rule norms to list out standards, but not editorialize on *why* those standards are important (i.e., "to ensure V2G systems can safely export power from EVs to the Utility's Distribution system"). This keeps it simpler and doesn't implicate whether a certain use case is required (in this case, whether export to the local area EPS/distribution system is needed).

Commented [ZW14]: All modern EVs have built-in inverters. VGIC recommends more appropriate and specific language.

Commented [ZW15]: There are two distinct certification pathways for V2G AC systems. Critically, **both** ensure compliance with the underlying IEEE 1547-2018 safety and grid interoperability requirements.

- 1)UL 1741 SC + SAE J3072. Some vendors may prefer this pathway, as it supports use cases requiring more EV<->EVSE interoperability.
- 2)UL 9741 + UL 1741 CRD. Some vendors may prefer this pathway, as it supports one-to-one matched pair certification. i.e., a specific charger with a specific vehicle, which may often be the case for a single-family home.

We respectfully urge NVE to adopt both pathways as options for customers/vendors to choose from. The market is relatively nascent, and limiting V2G AC interconnection to one pathway, despite both offering the core safety and grid interoperability requirements, would unnecessarily restrict the availability of these resources for customers.

Commented [ZW16R15]: Alternatively, this section could read:
"V2G AC EV systems shall either adhere to:
(1) UL 1741 SC for the EVSE and SAE J3072 for the EV; or
(2) UL 1741 CRD for DER systems as a UL 9741 matched-pair system."

This is likely a much more accessible way to list it for users of the Rule, and is consistent with the way these two pathways have been formatted in the Maryland rule.

backup operation and shall cease to discharge ~~before reconnecting to the Utility's Distribution System, prior to returning to normal operation Interconnected with the Utility's Distribution System.~~

Commented [ZW17]: Recommended for clarity.

9. Advanced Inverter Requirements

- a. Inverter-based Generators shall comply with IEEE 1547-2018 and UL 1741 SB requirements.
- b. All Advanced Inverters shall meet IEEE 1547-2018 Category B for normal operating performance and Category III for abnormal operating performance. Inverter-based Generators that cannot meet these requirements shall be reviewed on a case-by-case basis by the Utility and authorized for Interconnection or not at the Utility's discretion.
- c. The default settings identified within this section shall be used by all inverter-based systems regardless of size.
- d. The Utility may specify alternative site-specific settings on a case-by-case basis to preclude or reduce the Distribution System improvements that may be required as a direct result of the Interconnection or for other technical reasons unique to the location on the Utility's electric grid, and shall inform the Applicant and Producer, allowing reasonable time for mutual agreement of the parties prior to execution of the Interconnection and Operating Agreement.
- e. Default Settings
 - (1) Volt-VAR Settings
 - (a) This setting shall be enabled. All other reactive power control modes shall be disabled.
 - (b) Inverter-based Generators shall utilize a fixed reference voltage for this setting.
 - (c) Default settings shall conform to IEEE 1547-2018 Table 8.
 - (2) Volt-Watt Settings
 - (a) This setting shall be enabled.
 - (b) Default settings shall conform to IEEE 1547-2018 Table 10.
 - (3) Voltage Trip and Ride-Through
 - (a) Voltage trip settings shall conform to IEEE 1547-2018 Table 13.
 - (b) Voltage ride-through performance shall conform to IEEE 1547-2018 Table 16.
 - (4) Frequency Trip and Ride-Through
 - (a) Frequency trip settings shall conform to IEEE 1547-2018 Table 18.
 - (b) Frequency ride-through performance shall conform to IEEE 1547-2018 Table 19.
 - (c) Frequency droop performance shall conform to IEEE 1547-2018 Table 24.

10. Microgrid Requirements

- a. Requirements for a Microgrid shall be evaluated by the Utility on a case-by-case basis. Below are the defining characteristics for a Microgrid on the Utility's Distribution System:
 - (1) Is capable of Parallel Operation with the Utility's Distribution System or Islanded operation, and not solely as an emergency back-up resource;
 - (2) Is comprised of one or more Generator and/or ESD;
 - (3) Has clearly defined electrical boundaries;

- (4) Utilizes a Microgrid Interconnection Device at the PCC; and
- (5) Utilizes a control system (Master Microgrid Controller) to manage and dispatch resources as a single controllable entity.
- b. For a Microgrid comprised solely of inverter-based Generator technologies, at least one of the inverter-based technologies within the Microgrid shall be capable of Grid Forming operation when Islanded.
- c. Each Microgrid shall be designed to Interconnect with the Utility Distribution System, to operate without adversely affecting the operations of the electric grid and without presenting safety hazards to Utility personnel or Customers, and in the case of a Hybrid Microgrid, to operate without adversely affecting the operations of the Microgrid's participants.
- d. The Microgrid facilities and the Interconnection systems shall be in compliance with all applicable safety, power quality, and performance codes and standards of NFPA 70, IEEE, and the Utility's interconnection standards and procedures provided in this Rule.
- e. A Producer's Generating Facility may be part of a Customer Microgrid or be a participant in a Hybrid Microgrid and shall notify the Utility of its intention in the Interconnection Application.

11. Cybersecurity Requirements

- a. The Producer shall maintain good security practices regarding Generating Facility hardware and software in accordance with the Interconnection and Operating Agreement between the Producer and the Utility.
- b. Generating Facility hardware shall not be readily physically accessible to unauthorized persons (e.g., shall be locked or require some manner of authorized access) and strong passwords shall be assigned and maintained by the Producer for any device area network. Networked device default passwords and firmware reset shall be secured, abstracted, or made inaccessible.
- c. As applicable, equipment shall always be registered with the manufacturer.
- d. The Producer shall have and maintain security controls to protect the Utility's networks, systems, software, confidential information, and data, and to the maximum extent possible, shall adhere to the guidance provided in the latest published version of IEEE 1547.3-2022.
- e. Generating Facilities with communication, telemetry, and/or remote-control capability shall incorporate cybersecurity protections to prevent unauthorized access or disruption of the Utility's Distribution System. Such protections shall include cybersecurity policies, processes, and controls required for connectivity to the Utility systems, access to Utility data, or significant work affecting Utility systems.
- f. The Utility may establish and update its cybersecurity standards and procedures, which shall be incorporated into the Interconnection and Operating Agreement, and which may be updated from time to time and mutually agreed upon by the Producer and the Utility.

12. Communication Requirements

- a. The communication requirements in this section shall be between:
 - (1) the Utility and the individual Generating Facility's Advanced Inverter control, PCS, energy management system, or Master Microgrid Controller; or
 - (2) the Utility and the Generating Facility via a third-party aggregator; or
 - (3) other communication options mutually agreed to by the Producer and the Utility.
- b. Inverter-based Generators, including V2G EVs, shall adhere to all the following communication protocol requirements:

- (1) shall be capable of remote communications and adhere to IEEE 1547-2018, Clause 10;
- (2) software/firmware shall be capable of being updated locally or via remote communications;
- (3) transport level protocol shall be TCP/IP; and
- (4) shall be capable of supporting at least one of the following application-level protocols:
 - (a) IEEE 2030.5 (i.e., Smart Energy Profile 2.0 (SEP 2); or
 - (b) IEEE 1815 (i.e., Distributed Network Protocol 3 (DNP3)); or
 - (c) SunSpec Modbus; or
 - (d) other application-level protocols by mutual agreement of the Producer and the Utility.

~~e. V2G EVs with a built in inverter shall also adhere to standard SAE J2847/3 or be part of a UL 1741 SB Certified DER system.~~

~~d. V2G EV charger, charging station, and EVSE shall be capable of supporting at least one of the following protocols:~~

~~(1) IEEE 2030.5; and~~

~~(2) OpenADR (Open Automated Demand Response).~~

~~e.c.~~ Local communication ports on Advanced Inverters shall be made available at the request of the Utility or its designated third-party representative for the purposes of monitoring and control to support grid services, reliability tariffs or programs, or other tariffs or programs when such communication between Utility systems and the Advanced Inverters is required in accordance with any associated agreements between the Utility, its third-party representative, the Producer, and/or the Customer, or in accordance with any related Commission approved tariffs.

~~f.d.~~ Data Access. All operational data required by the Utility for safe, reliable, and tariff-compliant operation of the Distribution System shall be made available by the Producer or its vendors to the Utility, without additional fee, through the communication interface established under this Section. Required data shall include, but not be limited to, real and reactive power, voltage, frequency, state of charge, availability, and status of ride-through or trip functions, as applicable.

Commented [ZW18]: VGIC recommends striking these sections. Communication protocols between EV and EVSE or between EVSE and aggregators are out of scope for utility interconnection rules. All V2G EV or EVSE (i.e., AC or DC pathways) ensure compliance with IEEE 1547-2018, and IEEE 1547-2018 requires speaking either IEEE 2030.5, DNP3, or SunSpec Modbus. With this in mind, it is not necessary to define *additional* communications requirements. These additional requirements will likely create confusion in the marketplace and make it more difficult for customers to adopt these technologies.

VGIC recommends that NVE consider establishing communication requirements for EVSE in venues where it proposes funding for EVSE rebates and/or utility-administered managed charging or bidirectional charging programs.

H. METERING, MONITORING AND TELEMETRY

1. All Generating Facilities shall be metered in accordance with this Section H and shall meet all applicable standards of the Utility contained in the Utility's applicable Tariff Schedules and published Utility manuals dealing with Metering specifications.
2. The requirements in this Section H do not apply to Generating Facilities that are subject to separate metering requirements under the Utility's currently effective Tariff Schedules, as approved by the Commission, including but not limited to those established pursuant to NRS 704.766 to 704.775. Such Tariff Schedules, as may be amended from time to time, shall govern the Metering for these Generating Facilities.
3. The ownership, installation, operation, reading, and testing of Metering for Generating Facilities shall be by the Utility.
4. Telemetry and Metering at the PCC may be required in cases where less intrusive or more cost-effective options for providing the necessary Generating Facility output data are unavailable. In exercising its discretion to require Net Generation Metering, the Utility shall consider all relevant factors, including but not limited to:

- a. Data requirements in proportion to the need for information necessary to support relative Utility functions;
 - b. Producer election to install equipment that adequately addresses the Utility's operational requirements;
 - c. Accuracy and type of required Metering equipment to be equivalent and consistent (i.e., revenue grade) with purposes of collecting data;
 - d. The Generating Facility's size relative to the cost of the Metering/monitoring equipment;
 - e. Other means of obtaining the data (e.g. Generating Facility logs, proxy data etc.); and
 - f. Requirements under any Interconnection and Operating Agreement with the Producer.
5. PCC Metering
 - a. For purposes of assessing Utility charges for service, the PCC Metering shall be a bi-directional Meter so that power deliveries to and from the Producer's site can be separately recorded.
 - b. In order to accurately measure power flow in both directions, multiple Metering points may be required.
 - c. PCC Metering shall be equipped or designed to prevent reverse registration.
6. Telemetering
 - a. If the Net Nameplate Rating of the Generating Facility is 1 MW or greater, Telemetering equipment at the Net Generation Metering location may be required at the Producer's expense to allow the Utility and the Customer's scheduling coordinator to monitor such large Generators for their impact upon the Distribution System.
 - b. The costs for Telemetering, including equipment, installation, and any costs for leased telephone lines are separate and in addition to other Metering costs addressed in Section H.8 below.
 - c. If the Generating Facility is Interconnected to a Distribution System operating at a voltage below 10 kV, then Telemetering equipment may be required on Generating Facilities with a Net Nameplate rating of 250 kW or greater.
 - d. The Utility shall only require Telemetering to the extent that less intrusive or more cost-effective options for providing the necessary data in real time are unavailable.
 - e. Charges will be included in an applicable Interconnection and Operating Agreement.
7. If Utility-owned Metering is located on the Producer's Premises, the Producer shall provide, at no expense to the Utility, a suitable location for all such Metering.
8. Metering Equipment, including an appropriate revenue-quality Meter, as defined in Rule 1, shall be installed if required in the Producer's applicable standby rate schedule.
 - a. All service switches, Meter sockets, Meter enclosures, cutouts and similar devices, irrespective of voltage, required in connection with Net Generation Metering shall be furnished, installed, maintained and owned by the Producer at no cost to the Utility.
 - b. The Utility shall furnish, install and own the appropriate Meter or Meters used for Net Generation Metering. Such installation location shall be furnished by the Producer and approved by the Utility, and shall, at reasonable times, be accessible for reading, testing and maintaining the Meter.
9. Energy Storage Device Metering Equipment, including an appropriate Utility-owned, bi-directional, Interval Meter, as defined in Rule 1, shall be installed per applicable published Utility manuals and standards dealing with Metering specifications.

- a. All service switches, Meter sockets, Meter enclosures, cutouts and similar devices, irrespective of voltage, required in connection with Energy Storage Device Metering shall be furnished, installed, maintained and owned by the Producer at no cost to the Utility.
- b. The Utility shall furnish, install and own the appropriate Meter or Meters used for Energy Storage Device Metering, at its cost. Such installation location shall be furnished by the Producer and approved by the Utility, and shall, at reasonable times, be accessible for reading, testing and maintaining the Meter.

I. FACILITY OWNERSHIP AND COST RESPONSIBILITY

1. Facility Ownership

- a. Parallel Operation of Generating Facilities may require the installation of Interconnection Facilities or Distribution System improvements. The type, extent and costs of Interconnection Facilities and Distribution System improvements shall be consistent with this Rule and determined by the Utility through the technical review process and/or an Interconnection Study as described in Section F.
- b. Interconnection Facilities installed on the Producer's side of the PCC may be owned, operated and maintained by either the Producer or the Utility.
- c. Interconnection Facilities installed on Utility's side of the PCC and Distribution System improvements shall be owned, operated and maintained only by the Utility.

2. Facility Cost Responsibility

- a. A Producer shall be responsible for the reasonably incurred costs of the Optional Pre-Application Report (if requested) pursuant to Section E.1, the Application fee pursuant to Section E.2.c (Table 1) related to any required technical reviews pursuant to Section F, and any Interconnection Studies conducted pursuant to Section F.6 of this Rule solely to explore the feasibility and determine the requirements of Interconnecting a Generating Facility with the Utility's Distribution System.
- b. The Producer shall be responsible for all costs associated with the Interconnection Facilities owned by the Producer.
- c. The Producer shall be responsible for any costs reasonably incurred by the Utility in providing, installing, operating, or maintaining the Interconnection Facilities and Distribution System improvements required solely for the Interconnection of the Producer's Generating Facility with the Utility's Distribution System.
- d. The cost responsibility for any Interconnection Facilities and Distribution System improvements required solely due to the load requirements of an ESD (e.g., the ESD is charged by the Utility's Distribution System), EV charger, or EV charging station shall be as set forth in Rule 9.
- e. Should the Utility combine the installation of Interconnection Facilities or Distribution System improvements required for the Interconnection of a Generating Facility with improvements to the Utility's Distribution System to serve other Customers or Producers, the Utility shall not include the costs of such separate or incremental facilities in the amounts billed to the Producer.
- f. The costs for required Interconnection Facilities and Distribution System improvements shall be paid by the Producer pursuant to the provisions contained in the Interconnection and Operating Agreement. Where the type and extent of the Interconnection Facilities or Distribution System improvements warrant additional detail, the Producer and the Utility shall execute separate agreement(s) to more fully describe and allocate the parties' responsibilities for installing, owning, operating and maintaining the Interconnection Facilities or Distribution System improvements.

- g. Interconnection Facilities connected to the Utility's side of the PCC and Distribution System improvements shall be provided, installed, owned and maintained by the Utility at the Producer's expense.
- h. When a Producer wishes to reserve Utility-owned Interconnection Facilities or Distribution System improvements installed and financed as Special Facilities for the Producer but idled by a change in the operation of the Producer's Generating Facility or otherwise, the Producer may elect to abandon or reserve such facilities consistent with the terms of its Interconnection and Operating Agreement with the Utility. If the Producer elects to reserve idle Interconnection Facilities or Distribution System improvements, the Utility shall be entitled to continue to charge the Producer for the costs related to the ongoing operation and maintenance of the Special Facilities.

J. FACILITY INSTALLATION, INSPECTION, AND COMMISSIONING

- 1. Facility Installation
 - a. After executing the applicable agreements, the Utility and/or the Producer shall commence construction and installation of the Distribution System improvements or Interconnection Facilities that have been identified in the agreements using good faith efforts to meet schedules and fixed costs or estimated costs as appropriate.
 - b. Prior to construction, the Applicant may request, or the Utility may require, a pre-construction coordination meeting to review scope, switching or outage coordination, site access and safety controls. The Producer shall obtain all required permits and any AHJ clearances applicable to Producer-owned facilities. Work on Utility-owned facilities shall be performed only by the Utility. The parties shall use good faith efforts to coordinate schedules and minimize Customer impacts.
 - c. The Utility and the Producer shall confirm, in writing, the RPA established pursuant to Section F.1.e. for testing and commissioning per IEEE 1547-2018 requirements upon determination of Application completeness. Commissioning verification shall be performed at the agreed-upon RPA.
 - d. If required to complete the installation of a Generating Facility or associated Interconnection Facilities, the Utility may be required to perform a TDR to facilitate safe work on the Customer's service entrance equipment or to allow inspection, testing, or commissioning of the Generating Facility.
 - (1) If a TDR is required solely for modification of service entrance equipment (including installation or replacement of service panels), such work shall be performed and charged consistent with Rule 16.
 - (2) TDRs performed solely for modifications to Customer service entrance equipment pursuant to Rule 16 and not this Rule, shall be coordinated by the Utility as part of the facility installation process under Section J.1.a of this Rule. Such work shall not be deemed to initiate, extend, or toll of any timeline required by statute or regulation for the Utility's review of an Interconnection Application or issuance of Permission to Operate.
 - e. The Utility may issue a Temporary Interconnection Agreement solely to permit the physical connection of a Generating Facility to the Utility's Distribution System for limited purposes such as inspection, testing, coordination, and commissioning-only operation. A Temporary Interconnection Agreement shall not be construed as PTO and is valid only for a short, defined period necessary to complete commissioning.
 - (1) For Generating Facilities less than or equal to ~~25-30 kW~~ Net Nameplate Rating, Temporary Interconnection Authorization shall be issued automatically by the

Commented [ZW19]: See VGIC comment in Section F.1.b.(1)

Utility and shall allow commissioning-only, Non-Export operation for the minimum duration necessary to complete commissioning.

- (2) For Generating Facilities greater than ~~25-30~~ kW Net Nameplate Rating, the Utility shall specifically issue a Temporary Interconnection Authorization. The Producer shall notify the Utility prior to commissioning, and commissioning-only operation shall be authorized by the Utility within a reasonable timeframe and performed under coordinated conditions.
- (3) A Temporary Interconnection Authorization automatically expires upon the earlier of (i) issuance of PTO or (ii) Utility revocation.
- (4) Operation under a Temporary Interconnection Authorization remains subject to all safety and protection requirements of this Rule and all applicable permits required by the Authority Having Jurisdiction.

Commented [ZW20]: See VGIC comment in Section F.1.b.(1)

2. Facility Inspection

- a. Upon completion of installation and prior to Interconnection, the Producer shall notify the Utility and provide documentation including but not limited to:
 - (1) site plans and as-built one-line and/or three-line diagrams, clearly identifying the RPA;
 - (2) commissioning reports and documentation that may include verification of protection settings and inverter testing consistent with IEEE 1547-2018 and UL 1741 requirements (for Generating Facilities less than or equal to 1 MW Net Nameplate Rating, such reports shall consist of paperwork only);
 - (3) for Generating Facilities greater than 1 MW Net Nameplate Rating, functional Commissioning Testing and temporary energization in accordance with this Rule;
 - (4) verification of the Visible Break Isolation Device including documentation (including photos) confirming that the lockable, visible-open disconnect switch is installed in accordance with this Rule, Utility standards, and NEC;
 - (5) Certification of equipment including evidence that major components (inverters, relays, etc.) are Certified under UL 1741 SB/IEEE 1547-2018; and
 - (6) evidence of any relevant AHJ final inspection for Producer-owned facilities.
- b. Following receipt and review of the required documentation, the Utility shall perform a physical inspection of the facility that may include, but may not be limited to, verification of labeling, the Visible Break Isolation Device, protection wiring, Meter installation, and any other items necessary to confirm safe Interconnection consistent with applicable Utility standards.
- c. All inspections, testing, and verifications required under this Rule shall be performed at the RPA, as previously confirmed by the Utility and the Producer, to demonstrate compliance with any applicable technical and operational requirements, unless otherwise noted in this Rule.

3. Facility Testing

- a. The Producer shall be responsible for testing new Generating Facilities and associated Interconnection Facilities according to Attachment 2 and IEEE 1547.1-2020 to verify compliance with the safety and reliability provisions of this Rule prior to Parallel Operation with the Utility's Distribution System.
- b. The Producer's Generating Facility shall be authorized for Parallel Operation or Momentary Parallel Operation with the Utility's Distribution System only upon the issuance of PTO by the Utility.

- c. For non-Certified Equipment, the Producer shall develop a written testing plan to be submitted to the Utility for its review and acceptance prior to Interconnection. The testing plan shall, where applicable, incorporate the manufacturer's installation test procedures and conform to IEEE 1547.1-2020.
- d. The Utility may require a Utility-witnessed Commissioning Test. If required by the Utility, the Producer shall coordinate scheduling of such test with no less than five (5) business days' advanced notice and provide the Utility an opportunity to witness the test. Facility testing shall be conducted at a time mutually agreeable to the Producer and the Utility, with the party conducting the test providing the other party the opportunity to witness the tests.

4. Facility Certification and PTO

- a. The Producer shall be responsible for ensuring Certification of new Generating Facilities and associated Interconnection Facilities pursuant to Attachment 2, Section 2.
- b. For Net Metering Systems:
 - (1) within ten (10) business days of receipt of both written notice from the Applicant that the installation of the Net Metering System is complete and a copy of any required final inspection clearance from the AHJ, the Utility shall either:
 - (a) issue PTO; or
 - (b) provide to the Applicant a specific written description of the deficiencies that must be corrected prior to PTO.
 - (2) at the time the Utility issues PTO, the Utility shall provide to the Producer a copy of the fully executed Net Metering System agreement if it has not already done so.
- c. For non-Net Metering Systems, within ten (10) business days of receipt of notice from the Producer that installation of the Generating Facility is complete, and absent extraordinary circumstances, the Utility shall either:
 - (1) issue PTO; or
 - (2) provide the Producer with a written list of deficiencies to be corrected. Upon notice to the Utility by the Producer that all necessary correction have been made, the parties shall use good faith efforts to schedule any required re-inspection in a timely manner.
- d. The Producer's Generating Facility shall be authorized for Parallel Operation or Momentary Parallel Operation with the Utility's Distribution System only upon the issuance of PTO by the Utility. Issuance of PTO shall be contingent upon satisfactory completion of the requirements under this Section J, provision of all required documentation, and compliance with all applicable agreements.

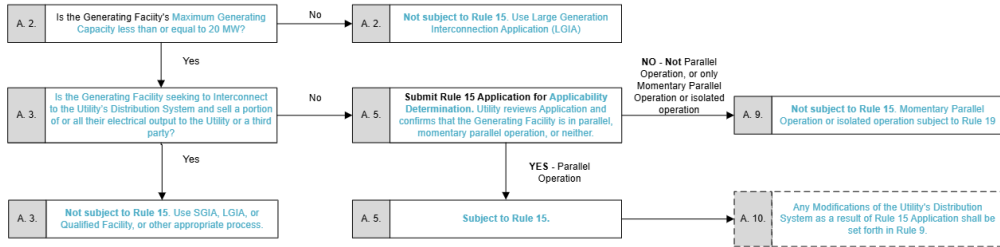
K. DISPUTE RESOLUTION

- 1. The Commission shall have initial jurisdiction to interpret, add, delete or modify any provision of this Rule or of any agreements entered into between the Utility and the Producer to implement this tariff ("the implementing agreements") and to resolve disputes regarding the Utility's performance of its obligations under its Tariff Schedules, the implementing agreements, and requirements related to the Interconnection of the Producer's Generating or Interconnection Facilities pursuant to this Rule.
- 2. Any dispute arising between the Utility and the Producer (individually "Party" and collectively "the Parties") regarding the Utility's performance of its obligations under its Tariff Schedules, the implementing agreements, and requirements related to the Interconnection of the Producer's facilities pursuant to this Rule shall be resolved according to the following procedures:

- a. The dispute shall be reduced to writing by the aggrieved Party in a letter ("the dispute letter") to the other Party containing the relevant known facts pertaining to the dispute, the specific dispute and the relief sought, and express notice by the aggrieved Party that it is invoking the procedures under Section K.2.
 - b. Within 45 calendar days after the Utility receives the dispute letter, the Parties' authorized representatives shall meet and confer to try to resolve the dispute.
 - c. If the Parties do not resolve their dispute within 45 calendar days after the date of the dispute letter, the dispute shall, upon demand of either Party, be submitted to resolution before the Commission in accordance with the procedures outlined in NAC 703.616 to 703.651.
3. Pending resolution of any dispute under this Section, the Parties shall proceed diligently with the performance of their respective obligations under this Rule and the implementing agreements, unless the implementing agreements have been terminated. Disputes as to the application and implementation of this Section shall be subject to resolution pursuant to the procedures set forth in this Section.

Attachment 1:

Applicability



Attachment 2: Summary Flowchart of Technical Review Process

Attachment 3:

Testing and Certification Criteria

1. INTRODUCTION

- a. This Attachment describes the test procedures and requirements for equipment used for the Interconnection of Generating Facilities to Utility's Distribution System. Equipment shall be subject to Type Testing, Production Testing, Commissioning Testing, and Periodic Testing consistent with IEEE 1547-2018, IEEE 1547.1-2020 and UL 1741-SB. These requirements ensure that the Generating Facilities will not adversely affect the Utility's Distribution System and that such facilities will cease to energize and provide power under abnormal conditions.

2. CERTIFICATION CRITERIA

- a. Equipment tested and approved (e.g. "Listed") by an accredited, nationally recognized testing laboratory (NRTL) as having met both the Type Testing and Production Testing requirements of IEEE 1547-2018 and UL 1741-SB shall be considered Certified Equipment
- b. Proof of Certification shall consist of an NRTL-issued certificate including:
 1. Effective date of Certification and serial/model numbers;
 2. The software version (if applicable);
 3. Test procedures procedure and revision;
 4. Laboratory accreditation; and
 5. Technical ratings, trip settings, and designations for Non-Exporting and Non-Islanding capability.
- c. The Applicant shall ensure that Certification documentation is available from the manufacturer, NRTL, or an authorized third party.
- d. Use of non-Certified equipment requires a site-specific testing plan and Utility approval consistent with IEEE 1547.1-2020.

3. TYPE TESTING

Type Testing shall be conducted in accordance with IEEE 1547.1-2020 and UL 1741-SB to verify compliance with Interconnection performance requirements including voltage and frequency ride-through, protective trip functions, reactive power, interoperability, and communications.

4. PRODUCTION TESTING

Production Testing shall be performed in accordance with IEEE 1547.1-2020 and UL 1741-SB as part of routine factory testing to verify each unit's performance.

5. COMMISSION TESTING

- a. Commissioning Testing shall be performed onsite in accordance with IEEE 1547.1-2020 to verify protective settings and functional performance upon initial Parallel Operation, after modifications affecting Interconnection functions, or when otherwise required.
- b. Testing shall be performed by a qualified engineer, factory-certified technician, or licensed electrician.
- c. The Utility may witness Commissioning Testing or require Certification by the installer of test completion results.
- d. For Certified Equipment, testing may be limited to verification of protective settings, anti-Islanding, and Non-Export Functions.
- e. For non-Certified Equipment, full Commissioning Testing per IEEE 1547.1-2020 is required or as mutually agreed to by the Producer and the Utility.
- f. At the completion of testing, the Producer shall confirm all devices are set to Utility-approved settings and provide documentation of such.

6. PERIODIC TESTING

Periodic Testing shall be conducted in accordance with IEEE 1547.1-2020 and manufacturer's requirements at intervals not to exceed four (4) years. The Producer shall maintain test records for Utility inspection. Batteries used for Protective Functions shall be inspected monthly and replaced or subject to a discharge test at least once every four (4) years.

7. SUPPLEMENTAL TESTING PROCEDURES

Supplemental Testing shall be conducted as required by IEEE 1547.1-2020 and UL 1741-SB, including but not limited to verification of Non-Exporting capability, in-rush current, surge withstand, and Synchronization performance.

8. Closing Provisions

Testing and Certification requirements under this Attachment shall be applied uniformly and consistent with IEEE 1547-2018, IEEE 1547.1-2020, and UL 1741-SB.